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**Desenvolvimento de um Dosímetro Pessoal Ativo e
Wireless para Radiologia de Intervenção**

**Development of a Wireless Active Personal
Dosimeter for Interventional Radiology**



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“He who would climb the ladder must begin at the bottom.”



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Dosimeter for Interventional Radiology**

Dissertação apresentada à Universidade de Aveiro para cumprimento dos requisitos necessários à obtenção do grau de Mestre em Engenharia Biomédica, realizada sob a orientação científica do Doutor Carlos Davide da Rocha Azevedo, professor auxiliar do Departamento de Física da Universidade de Aveiro, e do Doutor Pedro Manuel Mendes Correia (co-orientador), Investigador do Auxiliar do Departamento de Física da Universidade de Aveiro.

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Aos meus pais.

o júri / the jury

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Palavras-chave

Bremsstrahlung, calibração, cintilador plástico, dosímetro, fotomultiplicador de silício, radiologia de intervenção

Resumo

Este trabalho apresenta o desenvolvimento de um dosímetro pessoal ativo e *wireless*, de baixo custo, para radiologia de intervenção. Baseia-se num cintilador plástico BC-408 de $5 \times 5 \times 5 \text{ mm}^3$ acoplado a um fotomultiplicador de silício de $6 \times 6 \text{ mm}^2$, polarizado a 37 V. Foi projetada uma eletrónica de *front-end* em modo pulso, composta por um pré-amplificador de carga com modelação de pulso, um comparador, um circuito *peak-and-hold* e um ADC externo. É controlada por um microcontrolador ESP32-C3 com comunicação *Bluetooth Low Energy*. A cadeia de leitura tem um tempo morto de $17,8 \pm 0,4 \mu\text{s}$, correspondente a uma taxa máxima de $(56 \pm 1) \times 10^3$ contagens por segundo. Foi validada com um cristal de CsI(Tl) e fontes de ^{241}Am , ^{109}Cd e ^{57}Co , através do ajuste dos canais dos centroides das suas emissões às energias conhecidas. A resposta canal-energia mostrou-se linear ($R^2 = 0,999$) até ao canal 183 do ADC. O cintilador plástico foi calibrado em energia através do método do *endpoint* do espectro de *Bremsstrahlung* de um tubo de raio X, no qual este é obtido como a interseção do eixo das abcissas de um ajuste linear do limiar do espectro. Os espectros foram adquiridos entre 20 e 50 kVp, para cinco correntes no tubo, e o fotopico de 59,5 keV do ^{241}Am foi adicionado como ponto de calibração. Um ajuste linear ($R^2 = 0,989$) resultou num declive de $0,51 \pm 0,02$ canais/keV e numa ordenada na origem de 21 ± 1 canais. A resposta é linear entre 20 e 60 keV, com uma resolução em energia de $(29 \pm 5)\%$ a 59,5 keV. A frente de Compton do ^{57}Co foi também testada como ponto de validação, mas revelou-se pouco fiável a estas energias. A calibração em dose foi realizada em unidades de $H_p(10)$, segundo a norma ISO 4037, com as qualidades de radiação N-25, N-30 e N-40 e uma réplica do fantoma necessário. O espectro foi dividido em intervalos de energia, a cada um dos quais foi atribuída uma dose por contagem, e os pesos foram determinados pelo método dos mínimos quadrados. Foram testados três esquemas de divisão, tendo sido adotado o modelo de dois intervalos, com pesos de $93,5 \pm 11,4 \text{ pSv/contagem}$ para o intervalo de 20 a 25 keV e de $287,7 \pm 25,5 \text{ pSv/contagem}$ para o intervalo de 25 a 40 keV. O modelo reproduziu as taxas de dose de referência das três qualidades dentro da respetiva incerteza. As taxas de dose reconstruídas variaram entre 0,063 e $101 \mu\text{Sv/h}$ e apresentaram uma relação linear com a corrente no tubo ($R^2 \geq 0,998$). O protótipo final mede $6,3 \times 7,2 \times 2,5 \text{ cm}$, tem uma massa de 141 g e um custo estimado por unidade de aproximadamente 229 euros. Apresenta uma autonomia de 8,6 h a 100 eventos simulados por segundo.

Keywords

Bremsstrahlung, calibration, dosimeter, interventional radiology, plastic scintillator, silicon photomultiplier

Abstract

This work presents the development of a low-cost, wireless active personal dosimeter for interventional radiology. It is based on a $5 \times 5 \times 5 \text{ mm}^3$ BC-408 plastic scintillator coupled to a $6 \times 6 \text{ mm}^2$ silicon photomultiplier, biased at 37 V. A pulse-mode front-end was designed, comprising a charge-sensitive amplifier, pole-zero cancellation, a semi-Gaussian shaper, a comparator, a peak-and-hold circuit and an external ADC. It is controlled by an ESP32-C3 microcontroller with Bluetooth Low Energy communication. The readout reached a dead time of $17.8 \pm 0.4 \mu\text{s}$, corresponding to a maximum count rate of $(56 \pm 1) \times 10^3$ counts per second. It was validated with a CsI(Tl) crystal and ^{241}Am , ^{109}Cd and ^{57}Co sources, by fitting the centroid channels of their emissions against the known energies. The channel-to-energy response was linear ($R^2 = 0.999$) up to ADC channel 183. The plastic scintillator was calibrated in energy through the *Bremsstrahlung* endpoint method of an X-ray tube, in which the endpoint channel is taken as the x -intercept of a linear fit to the high-energy edge of the spectrum. Spectra were acquired from 20 to 50 kVp at five tube currents, and the 59.5 keV photopeak of ^{241}Am was added as a calibration point. A weighted least-squares fit ($R^2 = 0.989$) gave a slope of 0.51 ± 0.02 channels/keV and an intercept of 21 ± 1 channels. The response is linear between 20 and 60 keV, with an energy resolution of $(29 \pm 5)\%$ at 59.5 keV. The Compton edge of ^{57}Co was also tested as a validation point, but was found to be unreliable at these energies. The dose calibration was performed in terms of $H_p(10)$ following ISO 4037, with the N-25, N-30 and N-40 radiation qualities and a replica of the ISO slab phantom. The spectrum was divided into energy bins, each assigned a dose per count, and the weights were solved by least squares. Three binning schemes were tested, and the two-bin model divided at 25 keV was adopted, with weights of $93.5 \pm 11.4 \text{ pSv/count}$ for the 20 to 25 keV bin and $287.7 \pm 25.5 \text{ pSv/count}$ for the 25 to 40 keV bin. It reproduced the reference dose rates of the three qualities within their uncertainty. The reconstructed dose rates ranged from 0.063 to 101 $\mu\text{Sv/h}$ and were linear with tube current ($R^2 \geq 0.998$). The final wearable prototype measures $6.3 \times 7.2 \times 2.5 \text{ cm}$, has a mass of 141 g and an estimated component cost of approximately 229 euros per unit. It operates for 8.6 h at 100 simulated events per second.

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List of Abbreviations

ADC	Analog-to-Digital Converter	KERMA	Kinetic Energy Released in Matter
ALARA	<i>As Low As Reasonably Achievable</i>	LED	Light Emitting Diode
APD	Active Personal Dosimeter	LET	Linear Energy Transfer
BLE	Bluetooth Low Energy	LiPo	Lithium Polymer
CAD	Computer-Aided Design	MCU	Microcontroller Unit
CE	Compton Edge	MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
CSA	Charge-Sensitive Amplifier	NTC	Negative Temperature Coefficient
DAC	Digital-to-Analog Converter	OpAmp	Operational Amplifier
FB	Feedback	P&H	Peak-and-Hold
FWHM	Full Width at Half Maximum	PCB	Printed Circuit Board
HVL	Half-Value Layer	PMMA	Polymethyl Methacrylate
ICRP	International Commission on Radiological Protection	PS	Plastic Scintillator
ICRU	International Commission on Radiation Units and Measurements	PZC	Pole-Zero Cancellation
IR	Interventional Radiology	SiPM	Silicon Photomultiplier
ISO	International Organization for Standardization	SPI	Serial Peripheral Interface
		USB	Universal Serial Bus

Contextualization and Motivation

Interventional radiology (IR) refers to image-guided procedures performed under real-time X-ray visualization [1]. The number of these procedures has increased over the past two decades worldwide. Interventional radiologists and their teams are the most exposed group of medical radiation workers due to the prolonged use of fluoroscopy and the proximity required during interventions [2,3]. Currently, protection strategies rely on lead garments and structural shielding within the room [4]. Compliance with established occupational dose limits is monitored using passive dosimeters, which are worn either above or below the lead apron. These devices are read *a posteriori*, in the best-case scenario, once a month. As a result, there is no immediate feedback during procedures and exposure incidents cannot be prevented [5,6].

Active personal dosimeters (APDs)¹ offer a complementary solution by providing real-time auditory and/or visual feedback. The main benefit of these devices in IR is not to prevent exceedance of annual dose limits, which have been extensively studied and are rarely surpassed under standard conditions. Instead, APDs support the principle of keeping exposure *as low as reasonably achievable* (ALARA) [7–9]. These systems alert personnel to high-dose situations, promoting behavioural adjustments to reduce avoidable exposure. Adjustments may include reducing fluoroscopy time and frame rate, optimizing patient and operator positioning, and improving radiation shielding placement [10].

This dissertation explores the development of an APD system for IR, based on a plastic scintillator coupled to a silicon photomultiplier. The primary goal is to design, assemble and validate a pulse-mode front-end for this detector, in an architecture that can be replicated to monitor several anatomical regions, namely the eyes, neck, thorax and hands. Each unit is intended to be compact and fully wireless, so that it can be worn unobtrusively. Taking advantage of the tissue equivalence of plastic scintillators [11], a second goal is to evaluate their use for personal dosimetry, through the implementation and validation of methodologies for energy and dose calibration in the field of diagnostic radiology.

This document is structured in five chapters. Chapter 1 introduces the principles of IR and the physics of radiation interaction and dosimetry. It also characterises IR as a high occupational exposure environment, reviews APDs and their benefits, and presents the concept of the proposed dosimeter. The next chapter presents the preliminary studies and describes the design and validation of the front-end electronics. The third chapter covers the methodology used for the energy and dose calibration of the detector, along with the corresponding results and discussion. Chapter 4 shows the final prototype, resulting from successive design iterations, and reviews its channel response, wireless communication and autonomy. In the last chapter, the global conclusions of the work are presented and future work is discussed.

¹In this work, APD refers to active personal dosimeter and should not be confused with avalanche photodiode.

Introduction

1.1 Interventional Radiology

Interventional radiology (IR) is a subspecialty of radiology that uses fluoroscopy to perform diagnostic and therapeutic procedures through minimally invasive techniques [12]. Fluoroscopy uses X-rays and an image detection system to visualize internal structures and interventional tools in real time [13]. The conventional IR equipment configuration is illustrated in Fig. 1. The X-ray tube and the detector are mounted at opposite ends of a C-arm gantry, with the patient lying on the operating table between them. X-rays are produced in the tube by accelerating electrons from a heated cathode toward a target anode [1, 14]. The X-rays transmitted through the patient strike the detector, where they are converted into an electronic signal. This signal is processed into an image and displayed on a monitor to guide the intervention in real time. In the standard configuration, the X-ray tube is positioned beneath the patient table and the detector above, capitalising on the anisotropic distribution of scattered radiation to minimise occupational exposure [15, 16].

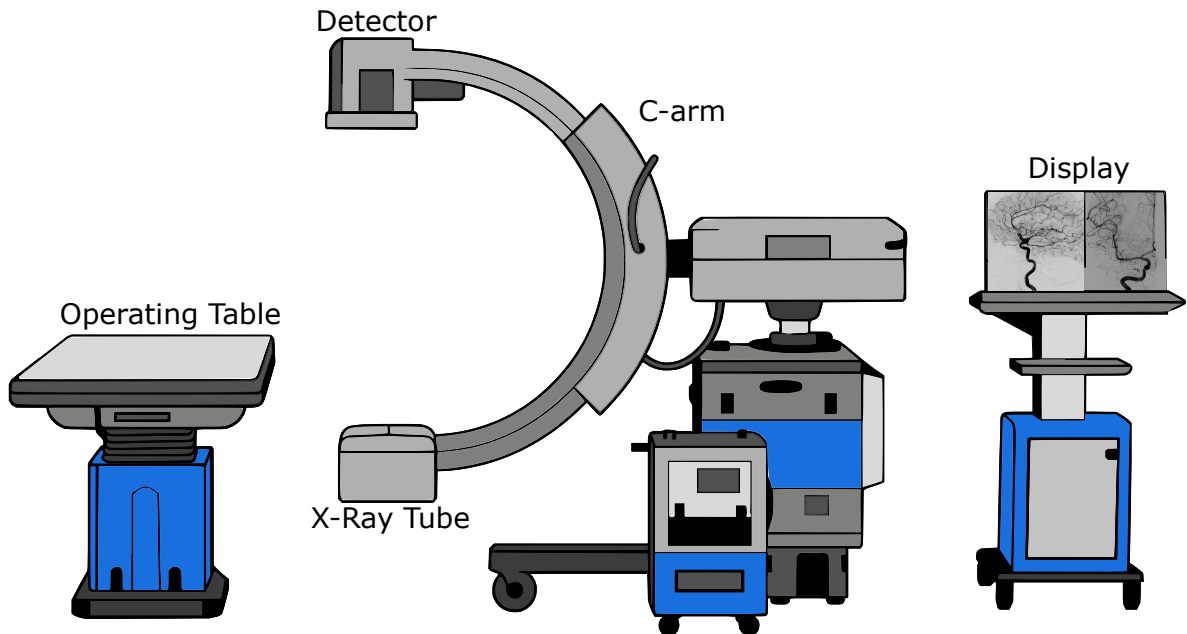


Figure 1: Interventional Radiology Setup. The C-arm is positioned in a posteroanterior projection (standard clinical practice) directing the most intense backscatter towards the floor, protecting above-knee radiosensitive organs [17]. The images shown on the display monitor were reproduced from Refs. [18, 19].

To maintain the perception of continuous motion, modern systems emit X-rays in short pulses synchronised with image acquisition, at rates of up to 30 frames per second. Since a single procedure may require thousands of images, the fluoroscopic detector operates in a considerably more sensitive mode than a radiographic detector. As a result, each image is produced with a dose several orders of magnitude lower than a radiograph [1]. Table 1 outlines common technical parameters used in IR procedures [20].

Table 1. Representative ranges of technical parameters used in IR procedures. Adapted from Ref. [20].

Parameter	Peak high voltage	Tube current	Inherent Al filtration	Cu filtration	Pulse duration	Pulse frequency	Scatter Energy
Range	60–120 kV	5–1000 mA	4.5 mm	0.2–0.9 mm	1–20 ms	1–30 Hz	20–100 keV

1.2 Photon Interaction with Matter

X-rays interact with matter through four physical mechanisms: Rayleigh scattering, the photoelectric effect, Compton scattering, and pair production. The contribution of Rayleigh scattering to overall interactions in the context of IR is negligible. Additionally, pair production requires photon energies exceeding 1.02 MeV, which are not present in diagnostic X-ray procedures [1]. Therefore, only the photoelectric effect and Compton scattering are considered in detail.

1.2.1 Photoelectric Effect

As shown in Fig. 2 (a), the photoelectric effect occurs when an incident photon transfers all of its energy to a bound electron within an atom, resulting in the ejection of the electron from its orbital. This process preferentially involves inner-shell electrons due to their greater electron density. The incident photon energy (E_0) is expended to overcome the binding energy of the electron (E_b), and any remaining energy is converted into the kinetic energy of the ejected photoelectron (E_{pe}), expressed as [1, 14]:

$$E_{pe} = E_0 - E_b, \quad (1.1)$$

The vacancy is filled by an outer electron, with the release of a characteristic X-ray or an Auger electron. When these emissions are reabsorbed within the detector, the total deposited energy equals E_0 , recorded as the full absorption peak, also called the photopeak. Due to the orbital quantization, the transitions are monoenergetic. However, and as represented, the peak has a finite width, set by the energy resolution of the detector, which is a property of the instrument [21].

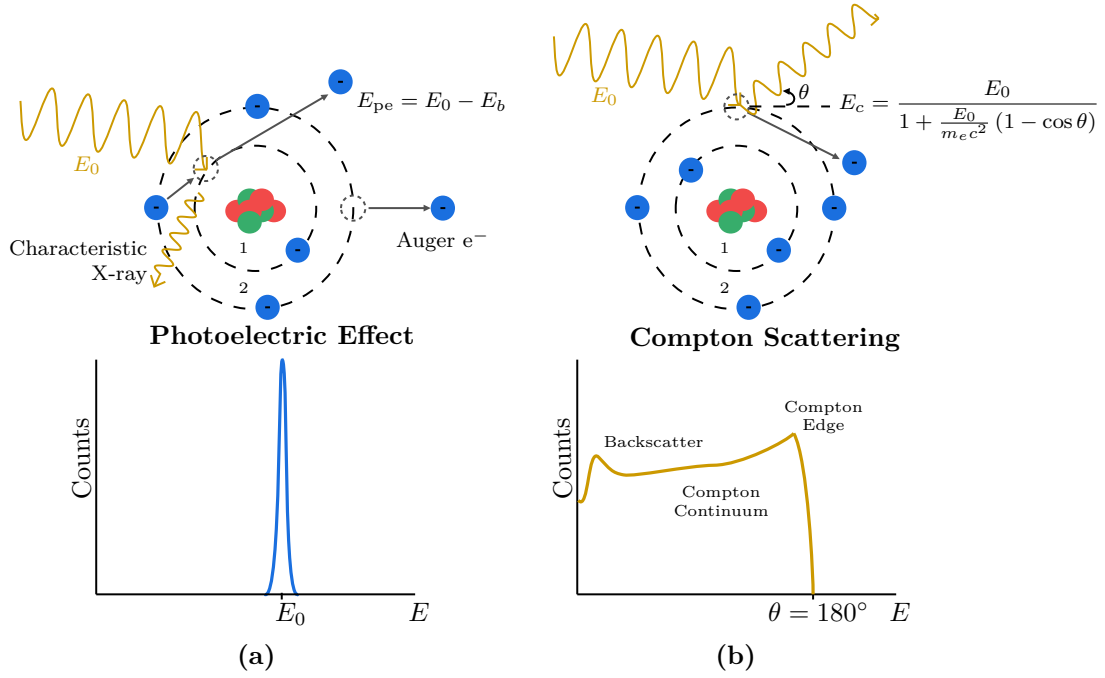


Figure 2: Schematic representation of the photoelectric effect (a) and Compton scattering (b), and the resulting energy spectrum.

1.2.2 Compton Scattering

Compton scattering is an inelastic interaction in which an incident photon interacts with a loosely bound electron, transferring part of its energy. As a result, the electron is ejected as a recoil (Compton) electron. In accordance with the conservation of energy and momentum, the photon is deflected at an angle θ , dependent on the energy transferred. The energy of the scattered photon E_c is described as [1]:

$$E_c = \frac{E_0}{1 + \frac{E_0}{m_e c^2} (1 - \cos \theta)} \quad (1.2)$$

Within the detector, all scattering angles are permitted, from 0° to 180° . This creates a Compton continuum, bounded by the two extremes. The maximum energy transfer to the electron occurs at $\theta = 180^\circ$, the Compton Edge (CE). At the same angle, a photon scattered in the surrounding material may enter the detector and be absorbed, forming the backscatter peak shown in Fig. 2 (b) [21].

In diagnostic and interventional radiology, the photoelectric effect is predominant at lower energies (<50 keV). Its probability increases with atomic number. This dependence provides contrast in X-ray imaging, as tissues of different composition absorb the beam differently. At higher energies, Compton scattering predominates in soft tissue, water and air. Its probability depends on the physical density of the material but not on the atomic number, since electron density is nearly uniform across soft tissues. The scattered photon retains most of the initial energy and remains sufficiently energetic to penetrate tissue, adding noise to the image. As a result, lower-energy X-ray beams produce higher contrast but also increase photon absorption in the patient, leading to a higher dose. Higher photon energies reduce patient absorption but

increase the contribution of Compton scattering, leading to greater noise and poorer contrast. In interventional planning, beam quality is therefore optimised to balance image quality and patient dose [1, 14].

1.3 Dosimetry Fundamentals

1.3.1 Physical Quantities

KERMA

When a beam of indirectly ionizing radiation, such as X-rays, interacts with matter, it transfers energy to charged particles within the medium. The total energy transferred (ε_{tr}) within a volume, V , is defined as [22]:

$$\varepsilon_{tr} = (R_{in})_u - (R_{out})_u^{\text{nonr}} + \sum Q, \text{ unit: [J]} \quad (1.3)$$

where $(R_{in})_u$ and $(R_{out})_u$ are the radiant energies of uncharged particles entering and leaving the volume, respectively, excluding secondary radiative emissions. $\sum Q$ represents changes in rest mass energy within V .

The dosimetric quantity *KERMA* (Kinetic Energy Released in MAtter) quantifies the initial kinetic energy transferred to charged particles, per unit mass, regardless of where and how that energy is spent [1, 22]:

$$KERMA = \frac{d\varepsilon_{tr}}{dm}, \text{ unit: } \left[\frac{\text{J}}{\text{kg}} \right] \text{ or [Gy]} \quad (1.4)$$

Absorbed Dose

Absorbed dose (D) accounts for the processes that follow the initial energy transfer, such as radiative losses, e.g. *Bremsstrahlung*. The total deposited energy (ε) extends the energy balance defined in Eq. 1.3 by excluding the energy lost through radiative processes and including all contributions from charged particles. Absorbed dose corresponds to the energy imparted per unit mass at a point of interest and quantifies the energy that remains in matter to produce radiation effects. It is expressed as [13, 22]:

$$D = \frac{d\varepsilon}{dm}, \text{ unit: [Gy]} \quad (1.5)$$

Exposure

Exposure (X) is the earliest defined dosimetric quantity and applies to photon interactions in air. It measures the amount of electric charge, dQ , produced by ionization in a given mass of dry air [22]:

$$X = \frac{dQ}{dm}, \text{ unit: } \left[\frac{\text{C}}{\text{kg}} \right] \quad (1.6)$$

Exposure is directly related to the collision component of *KERMA*, (K_{col}). K_{col} represents the fraction of energy transferred that is deposited through ionization and excitation, excluding radiative losses. In fact, a connection between X and K_{col} can be established through

the quantity $\overline{W}_{\text{air}}$, defined as the mean energy required to produce an ion pair in dry air ($\overline{W}_{\text{air}} = 33.97 \text{ J/C}$):

$$K_{\text{col, air}} = \overline{W}_{\text{air}} \cdot X = 33.97 \cdot X, \text{ unit: [Gy]} \quad (1.7)$$

In practice, $K_{\text{col, air}}$ is commonly referred to as *air KERMA* and is used to describe the output of an X-ray tube, as it can be measured directly in free air [13].

1.3.2 Protection Quantities

Radiation effects on humans vary due to differences in biological response and are classified as either deterministic or stochastic. Deterministic effects (Fig. 3 (a)) occur only above a threshold dose, beyond which their severity increases. Their thresholds are expressed in absorbed dose (Gy), and examples include cataracts and skin erythema. Stochastic effects (Fig. 3 (b)) have no threshold. Their probability of occurrence increases with dose, while their severity is independent of it. Cancer is the most relevant example. To reflect these probabilistic biological responses, additional weighting factors for radiation type and tissue radiosensitivity must be considered. Thus, two protection-related quantities, equivalent dose (H) and effective dose (E), are derived from absorbed dose [21, 23].

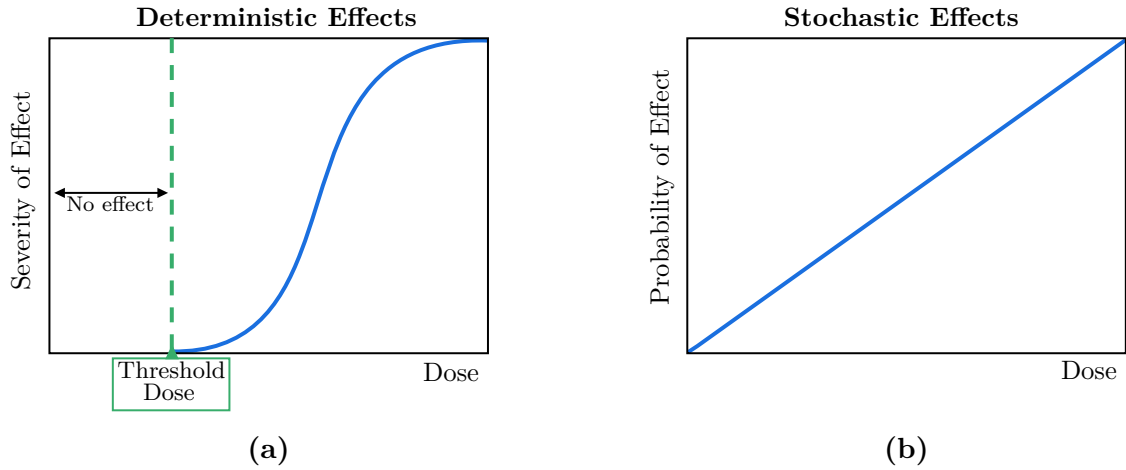


Figure 3: (a) Severity of deterministic effects as a function of dose. (b) Probability of stochastic effects as a function of dose [23].

Equivalent Dose

Different types of particles interact with matter in distinct ways, exhibiting different patterns of ionization along their paths [22]. This variation is described by the linear energy transfer (LET), a quantity that represents the rate at which energy is deposited along a particle's trajectory through a medium. Low-LET radiation (electrons and photons) produces ionization over longer paths, whereas high-LET radiation (α particles and protons) transfers energy more densely over shorter distances, enhancing its potential for biological damage per unit of absorbed dose [1]. To account for these differences, the International Commission on Radiological Protection (ICRP) introduced the concept of equivalent dose. This measure applies an empirical and dimensionless radiation weighting factor, w_R , to the absorbed dose, to reflect the relative biological effectiveness of different types of radiation [1, 21]:

$$H_T = \sum_R D_{T,R} \cdot w_R, \text{ unit: [Sv]} \quad (1.8)$$

The sievert (Sv) is dimensionally equivalent to joules per kilogram (J/kg) and gray (Gy), but specifically used for radiological protection to distinguish it from the physical quantity D [1]. Within radiology settings, for X-rays, $w_R = 1$, so equivalent and absorbed dose are numerically equal [24].

Effective Dose

Effective dose extends the concept of equivalent dose by incorporating tissue weighting factors, w_T , which quantify the relative contribution of specific organs and tissues to the overall radiation detriment. It is defined as the sum of the equivalent doses received by individual organs and tissues, each weighted by its respective factor [1, 21, 23]:

$$E = \sum_T H_T \cdot w_T, \text{ unit: [Sv]} \quad (1.9)$$

The values of w_T are normalised to unity and derived from epidemiological and radiobiological data, reflecting averaged population risk. Effective dose is therefore a comparative metric rather than an indicator of individual risk.

1.3.3 Operational Quantity

In operational dosimetry, equivalent and effective dose cannot be measured directly [23]. Personal dosimetry is the practice of using individual monitoring devices to estimate dose from occupational exposure. These devices are calibrated in terms of the personal dose equivalent, $H_p(d)$, defined as the dose equivalent in soft tissue at a depth d mm beneath a specified point on the body and is expressed in Sv. At 10 mm, $H_p(10)$ is used to estimate effective dose from deeply penetrating radiation and is the principal quantity for assessing whole-body exposure. Dosimeters intended to measure $H_p(10)$ are calibrated using reference radiation qualities defined in ISO 4037-1 [25]. During calibration, the dosimeter is mounted on a polymethyl methacrylate (PMMA) phantom, and the field is characterized in terms of *air KERMA*. For each radiation energy E and angle of incidence α , the International Commission on Radiation Units and Measurements (ICRU) provides conversion coefficients $h_{pK}(10; E, \alpha)$, which relate *air KERMA* to $H_p(10)$ [26]:

$$H_p(10) = h_{pK}(10; E, \alpha) \cdot K_{\text{air}}, \quad [\text{Sv}] \quad (1.10)$$

The same calibration approach applies to the assessment of equivalent dose, for which shallower depths are used. At 0.07 mm, $H_p(0.07)$ is used to assess dose to the skin from weakly penetrating radiation, and at 3 mm, $H_p(3)$ has been proposed specifically for the lens of the eye [23].

1.4 Occupational Exposure

As previously mentioned, at the photon energies typical of fluoroscopy (recall Table 1), the Compton effect predominates, degrading image quality and generating scattered radiation

throughout the room [1, 27]. This stray radiation is the main source of occupational exposure for staff [28]. From a dosimetric perspective, characterising the IR settings is inherently challenging due to the variability of parameters that influence radiation output and scatter distribution [29]. In general, exposure to staff is proportional to the dose received by the patient [15]. Tube voltage and current are adjusted based on patient size and anatomy during procedure planning, which in turn influence the amount of scatter produced [29, 30]. Beyond equipment settings, procedural complexity and operator technique contribute to variations in staff exposure. Advanced or prolonged interventions typically require longer fluoroscopy times, more image acquisitions, and non-standard projections [29]. Operator experience has been shown to significantly reduce exposure through more efficient techniques, reduced reliance on fluoroscopy, and better positioning practices [31].

To illustrate scatter behaviour in a real clinical context, Fig. 4 presents an isodose map, at chest height, generated during a fluoroscopically guided hip replacement procedure, and highlights the locations typically occupied by staff [32]. The image reveals an asymmetric scatter pattern, with notably higher intensities recorded opposite to the position of the C-arm. The referenced study also reported lower dose levels on the side opposite the patient table, where equipment provides partial shielding. Although the inverse square law does not perfectly apply in the clinical environment [33] due to scattering within the patient and surrounding structures, a general decrease in dose rate with increasing distance was still evident [32]. This reaffirms the protective effect of physical separation from the source [34].

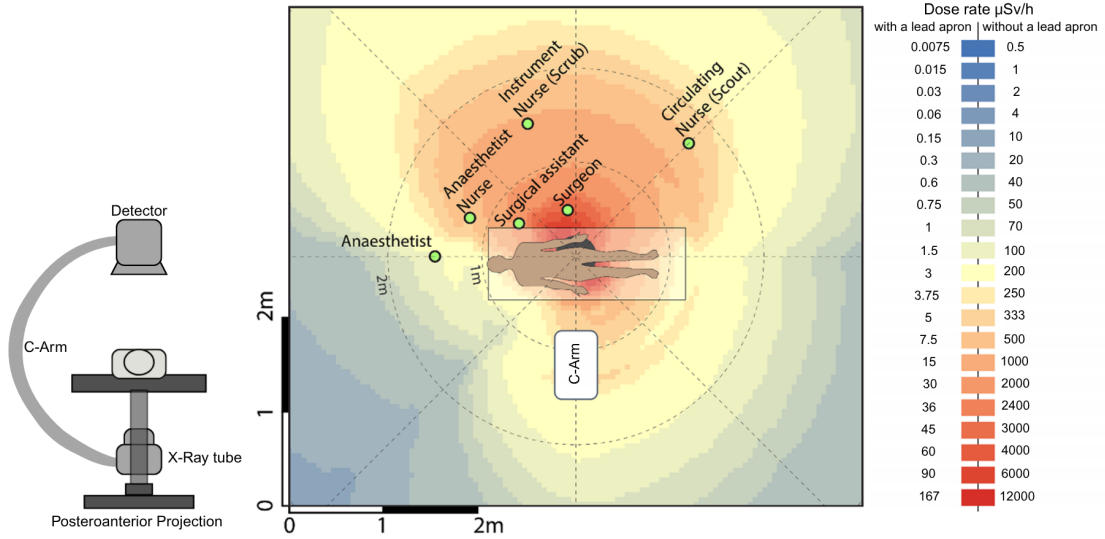


Figure 4: Isodose map of scatter radiation during a fluoroscopically guided hip replacement procedure. Acquisition parameters included 77 kVp, 20.6 mA, 32.1 s of fluoroscopy time, and a pulse rate of 15 fps. Dose rates were measured using an active dosimeter at multiple positions around the phantom and linearly interpolated to generate the map. Values incorporating shielding (right scale) were estimated using theoretical models. Measurements are documented at chest level. The image was adapted from Ref. [32]. Although the C-arm configuration was not explicitly reported, a posteroanterior projection (left icon) was inferred based on findings from Ref. [33].

Protection against cumulative exposure requires consistent application of shielding practices [4]. Shielding is implemented at multiple levels, including structural room design, fixed barriers, and mobile shields [35]. Still, personal protective equipment remains the most reliable and controllable form of protection. According to ICRP recommendations, all staff should wear lead aprons, along with thyroid shields, during fluoroscopic procedures [4, 35]. Most aprons have a lead thickness range of 0.35 to 0.5 mm, with 0.5 mm reducing scattered radiation by over 90% [4]. To be effective, aprons must fit properly, and facilities should be encouraged to invest in a variety of sizes to ensure proper coverage and comfort [36, 37]. Thyroid shields are manufactured with a lead equivalence of 0.5 mm and have been reported to reduce exposure to the gland by more than 50% [38]. Lead glasses are also available but remain underutilised. As the second most effective protective measure after ceiling-suspended shields, these glasses are available in various frame designs and lead thicknesses ranging from 0.35 to 0.75 mm [37].

The effectiveness of these protective measures is ultimately verified against the annual occupational dose limits recommended by the ICRP, summarised in Table 2. These limits are not an absolute safety threshold but rather boundaries beyond which cumulative exposures are no longer considered justified [23]. For instance, the E limit for radiation workers is set at 20 mSv per year, averaged over five years, with an upper limit of 50 mSv in any single year [24].

Table 2. ICRP occupational dose limits. Adapted from Ref. [24].

Protection Quantity		Annual Permissible Dose
Effective Dose		20 mSv per year averaged over 5 years (no single year exceeding 50 mSv)
Equivalent Dose	Lens of the Eyes	20 mSv
	Thyroid Gland	300 mSv
	Individual Organs (excluding the eye), Skin and Extremity	500 mSv

As previously introduced, currently, compliance with occupational dose limits in IR is verified through routine monitoring using passive dosimeters worn by staff over predefined time intervals. The most common devices are thermoluminescent dosimeters and optically stimulated luminescence dosimeters. Both operate by trapping energy when exposed to ionising radiation, which is later released as light when stimulated under controlled conditions. The intensity of the emitted light is proportional to the amount of absorbed radiation [13]. These devices measure the previously discussed quantity $H_p(10)$. While $H_p(10)$ can serve as an approximation of effective dose under conditions of uniform whole-body exposure, this does not reflect the typical environment in IR, as described [39]. When a single dosimeter is used, placement beneath the lead apron tends to underestimate E due to shielding, while positioning it above may overestimate it [29]. To improve accuracy, the ICRP recommends a dual dosimetry approach [40]. In this method, one dosimeter is worn underneath the protec-

tive apron between the waist and chest to record the attenuated exposure to shielded tissues (H^u). The second device is placed at collar level outside the apron to monitor exposure to unshielded regions (H^o). An effective dose estimate is then calculated by applying a linear combination of these two measurements [41]:

$$E = \alpha H^u + \beta H^o, \text{ unit: [Sv]} \quad (1.11)$$

However, there is currently no global consensus on the value of both weighting factors, α and β [41]. Over 14 different algorithms have been reported in the literature, most of which overestimate E . Despite this variability, double dosimetry remains more accurate than single-dosimeter methods [42].

The major limitation of passive dosimeters lies in their inability to provide dose information in real-time. As a result, opportunities for immediate corrections are lost, and staff may remain unaware of elevated exposure levels until the data can be reviewed retrospectively [5, 43]. In this context, active personal dosimeters (APDs) have emerged as a complementary solution. APDs are not intended to replace passive dosimeters, which remain the standard for legal dose documentation [8, 9].

1.5 Real-Time Dosimetry

APDs have been widely adopted in the nuclear industry as a standard safety practice [44], but only within the last 15 years has their use in IR been critically evaluated [8, 36, 44, 45]. Although several studies [20, 44, 45] explored the possibility of adapting existing devices for IR, it was Sánchez *et al.* [8], working as part of an early development initiative at Philips, who introduced the first APD specifically engineered for interventional use. Unlike existing APDs that provided exclusively auditory feedback, this prototype included a visual interface displaying real-time dose data to staff in the procedure room. It became the Philips DoseAware system [46], whose third generation, RaySafe i3, was introduced in 2017 [47].

RaySafe i3 features a silicon solid-state diode [48] integrated into a wearable badge, which transmits dose information wirelessly to a data collection unit at a one second rate. Dose values are displayed on an in-room monitor, with each badge represented individually through a colour-coded scheme (green, yellow, and red) corresponding to increasing dose rates, from 0.02 to 20 mSv/h [8]. The system operates over the N40–N150 reference radiation qualities of ISO 4037-1 [49] and has been shown to correlate well with legal-for-record passive dosimeters, typically overestimating $H_p(10)$ 5–15% [50]. The current limitations of the system are its cost and its anatomical coverage. Active devices of this type are expensive (> \$1000), which makes scaling to larger groups of radiation workers impractical [51]. Additionally, the badge is designed to be worn at chest level, and no system has yet been conceived to report the exposure to the eyes, hands or thyroid simultaneously, which are equally subject to annual dose limits.

1.5.1 Reported Benefits

The development and clinical integration of the APDs described above enabled structured evaluation of their practical impact in IR. Since their introduction, a number of studies have compared conditions with and without real-time feedback. A common finding is a general decrease in staff dose when feedback is available. This effect appears across various professional

roles, though the degree of reduction can vary. In a study led by Baumann *et al.* [52] dose reduction was observed in almost every staff group. Further analysis reveals a pattern seen across multiple studies, where the most significant reductions are recorded in the main radiologist, who is positioned closest to the patient. In some studies, this group was the only one to show a statistically significant change in radiation exposure [48, 52–54]. This may be explained by findings from Baumgartner *et al.* [48], which showed that procedural contexts associated with higher radiation exposure tend to demonstrate the most pronounced reductions following the introduction of APDs. This is consistent with reports that the effectiveness of these devices depends on baseline conditions. Operators who initially demonstrated poor radiation safety habits were able to significantly reduce their dose after the introduction of APDs through simple corrective actions. In contrast, those already practising good protection strategies had less room to improve and showed minimal change [50, 53, 55]. These baseline variations are also associated with the operator’s level of experience, as discussed in section 1.4, and were recently studied by Koch *et al.* [56]. A pronounced decrease in dose was registered among the least experienced group, while operators with over ten years of experience show, in this case, even a slight increase in dose. Experienced operators may rely on ingrained habits and confidence in their technique, whereas less experienced users appear to adjust more readily when confronted with live feedback [56]. It has been reported that younger professionals tend to show less concern for radiation safety [57]. These results suggest that APDs may be especially valuable as educational tools for junior staff, helping them build good radiation safety practices [56].

Consistent with these findings, James *et al.* [50], reported a general reduction in red events over time among different staff members. They argued that, while these events serve as prompts for immediate dose adjustment, they also function as a reflective tool for operators. This enables professionals to identify specific maneuvers, projection angles, and procedural choices associated with increased radiation exposure. Consequently, such practices can be replaced by lower-exposure alternatives and operators can adopt safer procedural habits more permanently. Ultimately, it also contributes to a reduction in radiation dose to the patient [50, 58].

Other staff members, including nurses and technicians, also benefited from dose reductions [50, 52, 54–56, 59]. In these cases, it was primarily attributed to individual awareness and simple adjustments, such as changes in positioning. Some studies further emphasised the collaborative value of these systems, noting that when all staff members wear APDs with visible displays, teams can collectively monitor exposure. This practice creates an environment of mutual accountability, where individuals not only adjust their own behaviour but also remain aware of their colleagues’ exposure, promoting safer team dynamics [50, 52, 53].

1.5.2 Proposed Dosimeter Concept

In this context, this work proposes a low-cost wearable active personal dosimeter for IR. Its conceptual design is shown in Fig. 5. The system consists of a compact wearable unit, which may be worn over or under the protective apron and integrates the full detection, pulse processing and transmission chain. Each stage is described in detail in the following chapter. The target energy range is 20 to 100 keV, corresponding to the energies of the scattered field in IR (recall Table 1).

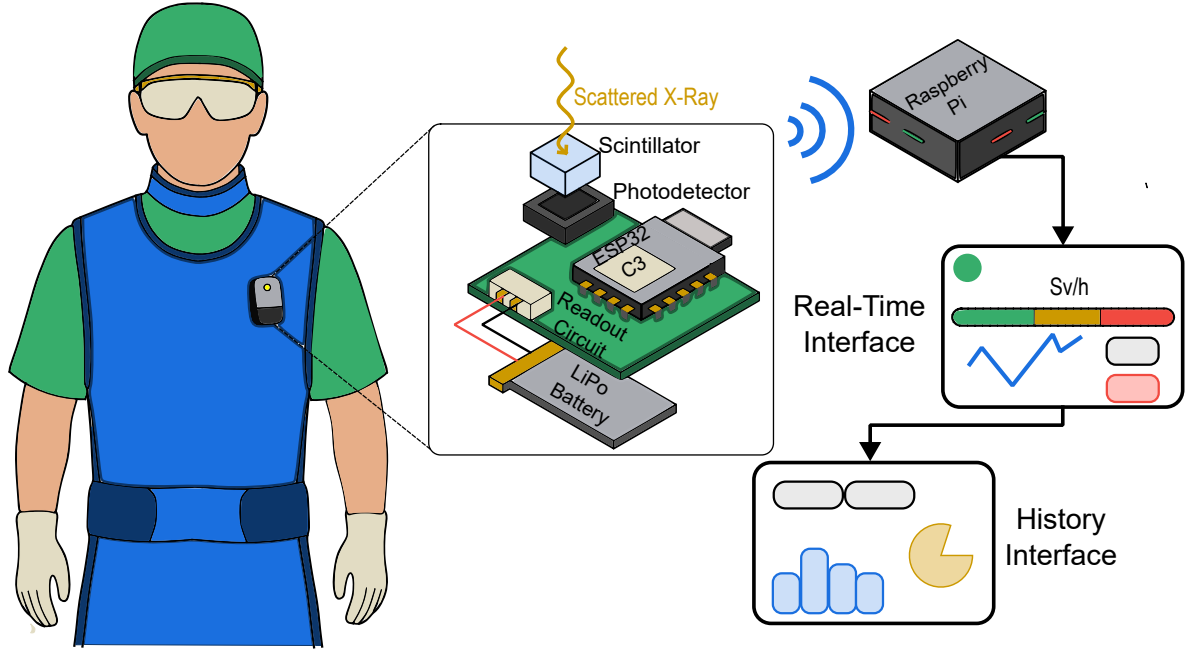


Figure 5: Conceptual design of the proposed dosimeter. The wearable unit (exploded view) may be worn over the protective apron and comprises the scintillator, photodetector, readout circuit, MCU and LiPo battery. Data are sent to a Raspberry Pi, which displays the real-time and history interfaces.

Scintillation Detection

The detection of the scattered radiation is based on a plastic scintillator (PS). PSs are organic materials that emit photons in the near ultraviolet to visible range when excited by ionising radiation. Their main advantage is their tissue equivalence. Their effective atomic number is also lower than that of the materials used in passive dosimeters or in solid-state diodes, like the RaySafe i3. The energy absorbed in the scintillator therefore approximates the dose to soft tissue without requiring further corrections [11]. However, as a drawback, the same tissue equivalence implies that, as in soft tissue, Compton scattering predominates over the photoelectric effect at the photon energies of the scattered field. As a result, the acquired spectrum reduces to a Compton continuum without photopeaks, and the energy resolution of these detectors is poor (recall Section 1.2). Additionally, below approximately 125 keV, the light output of PSs is non-proportional due to ionisation quenching. Slow (lower energy) electrons ionise the medium more densely, and part of the excitation is dissipated without the emission of light [60]. The Compton electrons produced in the PS in this work fall below this threshold. The pulse height of the detected events may therefore not be a linear measure of the deposited energy.

The scintillation light is converted into an electrical signal by a silicon photomultiplier (SiPM). A SiPM consists of a parallel array of avalanche photodiodes operated in Geiger mode. Each cell is reverse biased above its breakdown voltage, so a single absorbed photon is sufficient to trigger a self-sustaining avalanche. The photons produced by a scintillation event fire many cells simultaneously, each of which contributes one identical pulse. The summed output current is proportional to the number of fired cells, and therefore to the energy deposited in the PS [21].

Readout and Transmission

The dosimeter is intended to operate in pulse mode, where each interaction is registered as an individual event. The corresponding pulse is conditioned by a readout board, which preserves the amplitude and thereby the energy deposited in each event. By accounting for the energy of individual interactions, rather than integrating total charge, the accuracy of the dose calculation can be improved.

The chain is controlled by a microcontroller unit (MCU) with wireless communication capabilities. The ESP32-C3 was selected for being compact, lightweight and low-cost. It supports 2.4 GHz Wi-Fi and Bluetooth[®] Low Energy (BLE) 5 [61], the latter being the adopted communication protocol. Unlike Wi-Fi, BLE establishes a direct link with the receiver and requires no external router, reducing system complexity. In addition, BLE 5 offers extended range and increased data throughput relative to previous versions, meeting the requirements of continuous dose transmission. The unit is powered by a rechargeable lithium polymer (LiPo) battery, the AKYGA LP103040 [62], whose 1200 mAh capacity supports uninterrupted operation over extended periods. The selected Seeed Studio XIAO ESP32-C3 [63] board already incorporates a charging circuit for the battery via its USB-C port. Accordingly, the front-end electronics operate at a 3.3 V logic level.

Data Processing and Interface

To preserve battery life, the wearable unit does not compute dose. The MCU accumulates the amplitude information of the detected events and transmits it via BLE at a one-second rate to a Raspberry Pi. Since the Raspberry Pi is permanently powered, it handles all data processing, from pulse amplitude to energy and, ultimately, to dose. A real-time interface then displays the dose rate with a complementary colour-coded scale, updated at the same one-second rate. The scale is replicated on an LED integrated in the wearable unit, with the corresponding colour level sent back to the badge via BLE. All measured data are also stored in an internal database, from which a second interface builds retrospective summaries of the recorded exposure.

Prototype Development

2.1 Preliminary Studies

2.1.1 Experimental Setup

The concept proposed in the previous chapter relies on the premise that a PS read by a SiPM produces a measurable signal under X-rays in the relevant energy range. To become familiar with the detection chain, a first sensing element (a SiPM coupled to a PS) was assembled with the components available at the beginning of the project.

The SiPM was a Broadcom AFBR-S4N22P014M [64], with a $2 \times 2 \text{ mm}^2$ active area. Its breakdown voltage lies between 32 and 33 V at 25 °C, and its photon detection efficiency peaks at 63% at 420 nm, for a 12 V overvoltage. The scintillator was a BCF-12 fibre [65], whose emission peaks at 435 nm. A 5 mm length was cut from a 2 mm diameter fibre, and both ends were polished. The SiPM was mounted on a small PCB giving pin access to its anode and cathode, and optical silicone grease [66] was applied between the SiPM and the fibre. As shown in Fig. 6 (a), the assembly was placed in a 3D-printed enclosure sealed with black tape to prevent light from entering. Current measurements were performed with a Keithley 6487 picoammeter [67], in the configuration shown in Fig. 6 (b). The same instrument also biased the SiPM. Two configurations were measured: the SiPM alone and the SiPM coupled to the fibre. In both, X-rays were produced by the Apogee 5500 Series tube (Oxford Instruments) used throughout this work, a 50 W tube that operates from 0 to 50 kV with a maximum current of 1.0 mA [68].

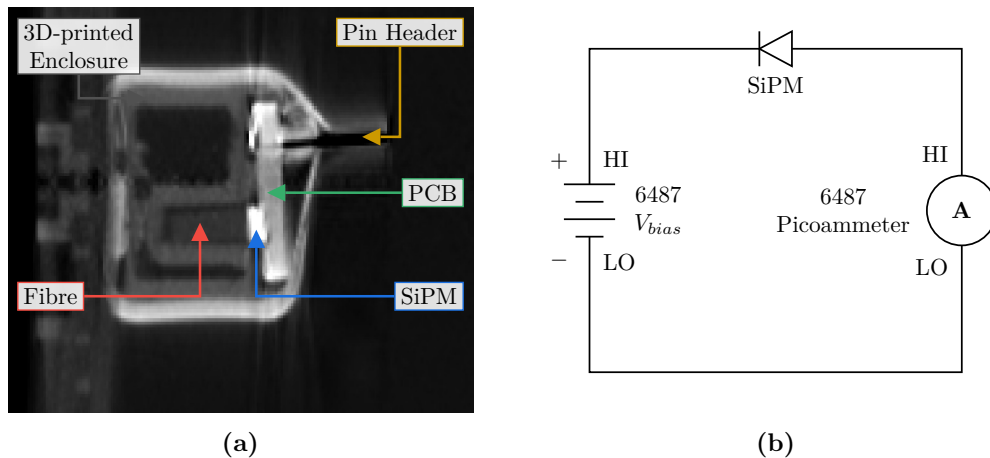


Figure 6: Preliminary measurement setup. (a) Radiograph of the SiPM and fibre assembly inside the 3D-printed enclosure. Tube voltage and current were not recorded. (b) Bias and current measurement circuit with the Keithley 6487.

2.1.2 Results

Fig. 7 shows the mean current as a function of the SiPM bias voltage, for both configurations and all tested tube voltages. Below 30 V, the current remains at the nanoampere level in all cases. It is leakage current, since the electric field is not sufficient to trigger avalanches, even under irradiation. Beyond the grey band, the SiPM operates in Geiger mode and the dark current rises. In this region, with the tube off, both configurations converge, so the baseline is the SiPM's own dark current. In the presence of radiation, the measured signal is dominated by the fibre. Although, in all cases, the interaction of X-rays in the silicon raises the measured current, the signal due to radiation increases by at least one order of magnitude with the fibre. The least distinct case is 15 kV, consistent with the lower photon fluence at this tube voltage. Direct interaction in the SiPM is therefore a minor contribution, and coupling the SiPM to a PS produces a distinct signal.

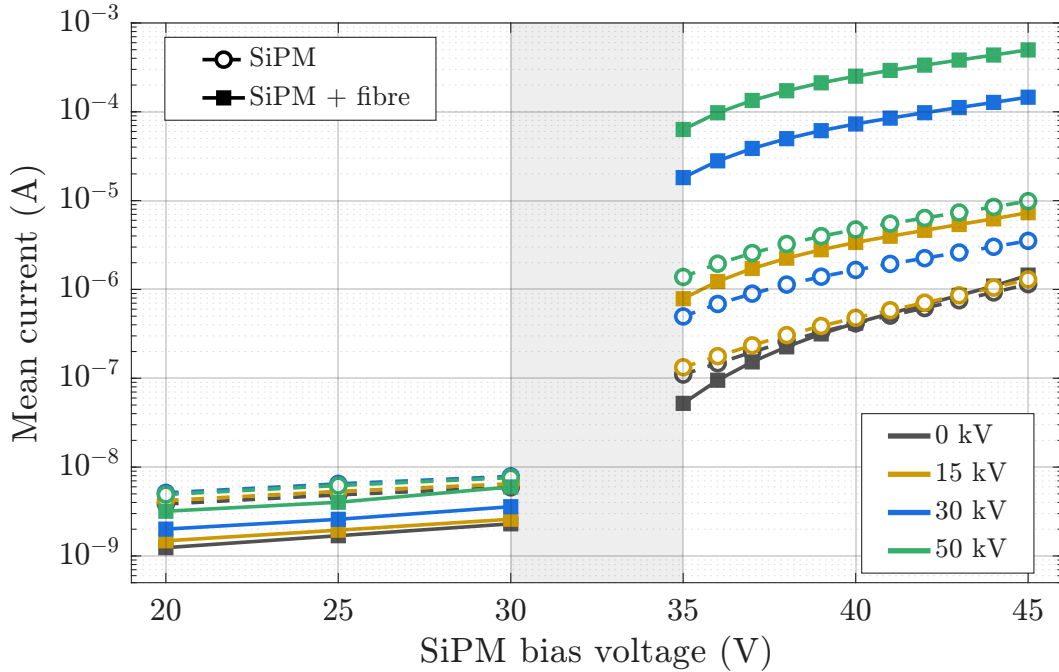


Figure 7: Mean current as a function of the SiPM bias voltage, for the SiPM alone and coupled to the fibre, at tube voltages of 0, 15, 30 and 50 kV and a tube current of 1 mA.

2.2 Hardware Overview

With the detection principle validated, the PS selected for the prototype was a $5 \times 5 \times 5$ mm³ BC-408 cube. Unlike the elongated fibre, the cube offers a larger interaction volume and a cross-section less dependent on the direction of incidence [69]. BC-408 consists of a polyvinyltoluene base doped with fluorophores that shift the emission to a peak wavelength of 425 nm [70], closer to the peak sensitivity of the SiPM. The photodetector was also replaced by the AFBR-S4N66P014M [71], whose 6×6 mm² active area covers the full cross-section of the PS, maximising light collection.

Since each microcell fires at most once per event, the number of fired cells must remain well below the 22 428 microcells of the SiPM [71]. With a light yield of 10 000 photons/MeV [70],

BC-408 produces about 1 000 photons at 100 keV, the upper limit of the target range. Even with complete light collection and the 63% peak photon detection efficiency of the SiPM, this corresponds to at most about 630 fired cells, fulfilling this requirement.

2.2.1 SiPM Power Supply

Although gain increases with overvoltage, so do the dark count rate and crosstalk probability, which in excess mask the low-energy end of the spectrum. For the 20 to 100 keV range, an overvoltage of about 4 V provides enough gain for these events to sit above noise levels. The operating voltage was therefore set to 37 V. The bias is generated by a MAX1932 [72], whose output is programmed by the MCU through a Serial Peripheral Interface (SPI). The MCU writes an eight-bit code to the internal Digital-to-Analog Converter (DAC) of the power supply.

A fixed 37 V output does not account for the variation of the breakdown voltage with temperature, which increases at 30 mV/°C according to the SiPM datasheet [71]. Since gain depends on the overvoltage, a constant bias means the gain decreases as temperature rises. The same deposited energy then produces a different pulse amplitude, affecting the energy calibration and the derived dose. To compensate for this, a negative temperature coefficient (NTC) thermistor was added to the feedback (FB) network, as recommended in the MAX1932 datasheet [72] and shown in Fig. 8.

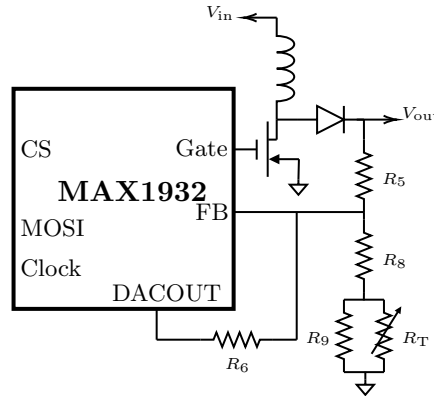


Figure 8: MAX1932 feedback network with the NTC thermistor R_T for temperature compensation of the SiPM bias. Pin labels on the left correspond to SPI connections.

The total resistance between the feedback node and ground is [72]:

$$R_{\text{total}}(T) = R_8 + \frac{R_9 R_T(T)}{R_9 + R_T(T)}, \quad (2.1)$$

The output voltage is therefore [72]:

$$V_{\text{OUT}} = 1.25 \left(1 + \frac{R_5}{R_{\text{total}}(T)} \right) + \frac{R_5}{R_6} (1.25 - V_{\text{DAC}}), \quad (2.2)$$

where V_{DAC} is the DAC output voltage. The compensation condition requires the derivative of Eq. 2.2 to equal the temperature coefficient of the breakdown voltage around room

temperature:

$$\frac{dV_{\text{OUT}}}{dT} = \frac{1.25 R_5}{R_{\text{total}}^2} \left(\frac{R_9}{R_9 + R_T} \right)^2 \frac{\beta}{T^2} R_T = 30 \text{ mV}/^\circ\text{C}, \quad (2.3)$$

where β is the thermistor constant and T the absolute temperature. Solving this condition together with Eqs. 2.1 and 2.2 and the component selection equations of the datasheet [72] yields $R_5 = 91 \text{ k}\Omega$, $R_6 = 27 \text{ k}\Omega$, $R_8 = 2.4 \text{ k}\Omega$, and $R_9 = 910 \Omega$.

Finally, the DAC code was set so that the output is 37 V at the operating temperature of the board.

2.2.2 Front-End Electronics

The front-end electronics perform the pulse-mode readout of the SiPM response. The architecture developed for this purpose is represented by the simplified block diagram in Fig. 9, and includes three stages. Pulse shaping integrates and filters the SiPM current into a voltage pulse whose amplitude is proportional to the collected charge. Pulse discrimination then compares this pulse against a reference threshold, rejecting noise, and retains its peak amplitude. Pulse sampling digitises the held value through an Analog-to-Digital Converter (ADC) and transmits it to the MCU. Each stage is detailed in the following sections.

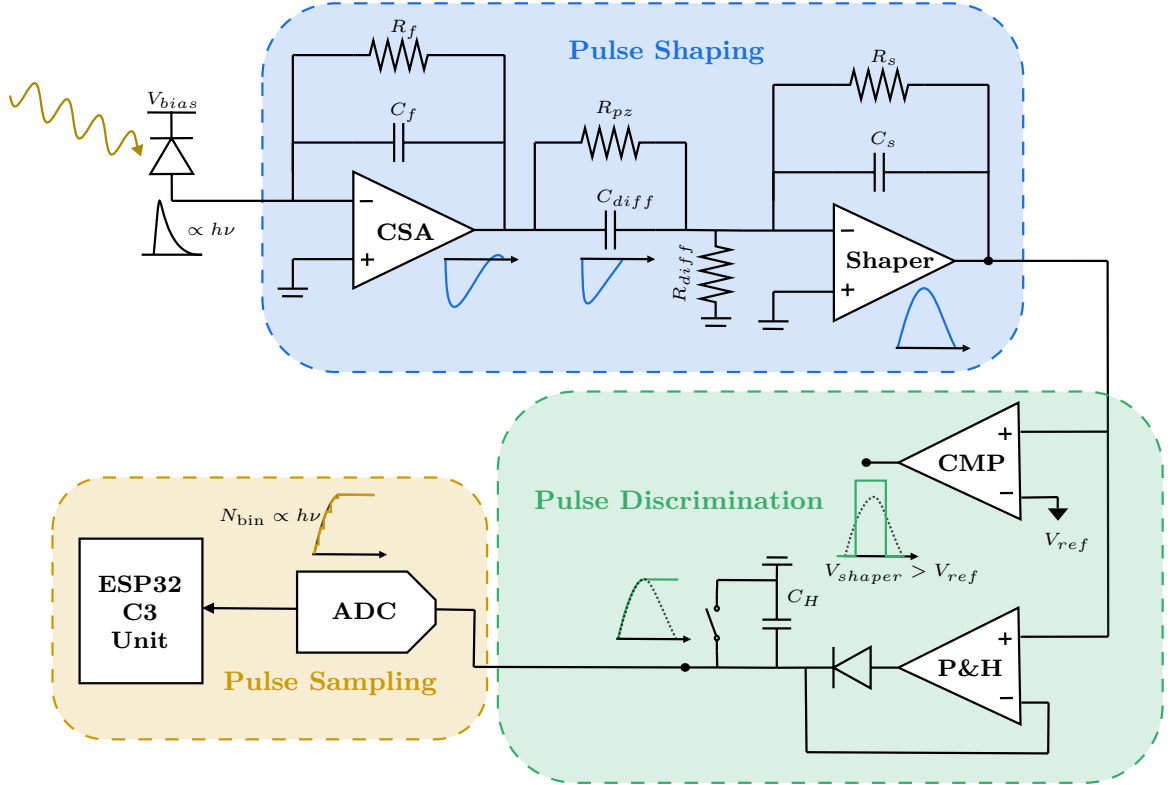


Figure 9: Architecture of the proposed dosimeter.

The front-end electronics were assembled on a printed circuit board (PCB), which also integrates the ESP32-C3, the MAX1932 bias supply and the SiPM described previously. The prototype test board is shown in Fig. 10 (a). The scintillator and the SiPM are not soldered

directly on the board. Instead, it is the same arrangement of Fig. 6 (a), in a light-tight printed box connected to the readout chain through a pin header. This allows the detector elements to be replaced without reworking the board, so that different configurations can be evaluated with the same electronics. All measurements presented in this work were performed with three sealed radioactive sources, ^{241}Am , ^{109}Cd and ^{57}Co , selected because their emissions fall within the energy range of interest. Further information on their activities can be found in Appendix A. A test box was 3D printed to align each source with the active area of the photosensitive housing, as shown in Fig. 10 (b). The box additionally reduces the handling of the sources and allows a greater distance to be maintained during testing, contributing to a lower radiation exposure.

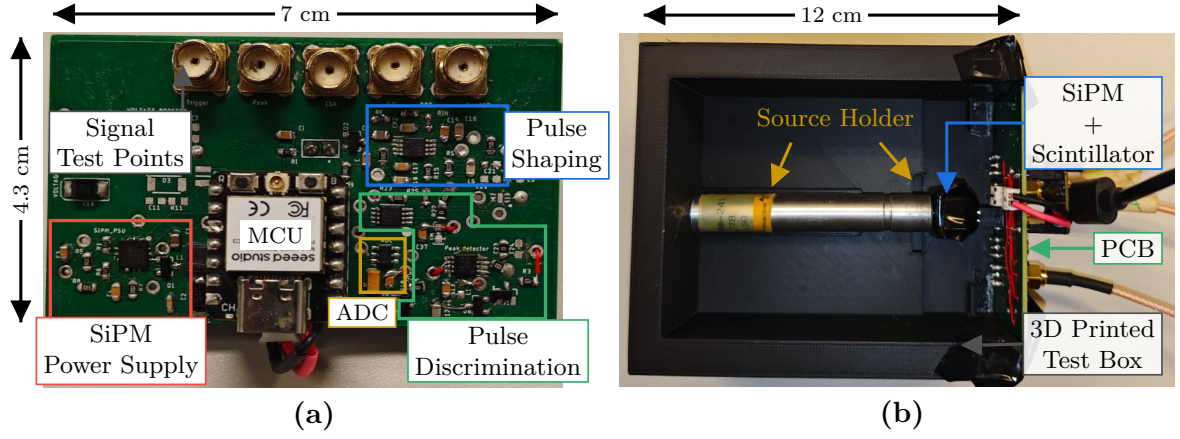


Figure 10: Prototype test board. (a) PCB with the main functional blocks identified, together with the signal test points used for characterisation; (b) assembled prototype inside the 3D-printed test box, showing the sealed source (^{241}Am) and source holders aligned with the scintillator and SiPM.

2.3 Pulse Shaping

The current pulse produced by the SiPM delivers a quantity of charge, Q , that must be converted into a measurable voltage signal in the preamplifier [73]:

$$Q = \int I(t) dt \quad (2.4)$$

To achieve this, a Charge-Sensitive Amplifier (CSA) was implemented. The CSA is widely used as a preamplifier since the rise time of the output pulse depends only on the charge collection time in the detector, independent of both the detector and the preamplifier input capacitances [21]. It is an inverting integrator configuration designed around a low-noise operational amplifier, such as the LT6203 [74], that was chosen accordingly. The feedback loop is closed by a capacitor C_f in parallel with a resistor R_f [21, 75].

In this circuit, the current flows through C_f , allowing it to store charge and producing an output voltage proportional to Q . R_f acts as a reset mechanism, providing a discharge path for C_f , in order for new pulses to be detected. The resulting output voltage can be written as [73, 76]:

$$V_{\text{out}} = -\frac{Q}{C_f} e^{-t/\tau_f} \quad (2.5)$$

where $\tau_f = R_f C_f$ is the discharge time constant. For total charge integration, this constant must be longer than the duration of the SiPM current pulse.

The charge gain of the CSA, G , depends exclusively on C_f and is therefore [75]:

$$G = \frac{V_{\text{out}}}{Q} = -\frac{1}{C_f} \quad (2.6)$$

To preserve the pulse amplitude, which carries the energy information of the detected events, additional signal conditioning stages were implemented. Given the high event count rate expected in IR, a Semi-Gaussian shaper was adopted in order to prevent baseline drift and relax the timing demands of the following stages. It consists of a CR differentiator followed by n RC integrator stages. Increasing n produces a more symmetric output pulse, which returns to baseline faster but introduces a longer peaking time [21]. For this work, a simple CR-RC ($n = 1$) network was implemented, with equal differentiation and integration time constants.

The finite decay time constant of the CSA causes an undershoot at the shaper output [77]. If the signal has not recovered before the next event arrives, the subsequent pulse is superimposed on this undershoot, reducing its measured amplitude. To mitigate this effect, a Pole-Zero Cancellation (PZC) configuration was implemented. C_{diff} and R_{diff} form the CR differentiation network. A resistor R_{pz} , placed in parallel with C_{diff} , introduces a zero that compensates the pole produced by the CSA. Proper cancellation occurs when the following condition is satisfied [21, 78, 79]:

$$R_{\text{pz}} C_{\text{diff}} = R_f C_f \quad (2.7)$$

The RC stage was implemented as an inverting low-pass filter, similar to the CSA configuration, with a feedback capacitor C_s in parallel with a resistor R_s . As in the CSA, the gain depends on the feedback capacitor, so the value of C_s controls the gain of the stage. The value of R_s is then chosen so that $R_s C_s$ equals the common time constant fixed by the previous stages.

The gain of the front-end electronics was set empirically. As mentioned in Section 2.2.1, the SiPM bias was fixed to keep the dark count rate at a low amplitude without masking the low-energy end of the spectrum. The gain must be high enough that real pulses are resolved above this level. It must also be low enough that the full energy range of interest, up to 100 keV, remains below saturation. The gain therefore places the spectrum in the lower portion of the available 3.3 V amplitude range. Within this portion, enough channels must remain available to sample these energies. Following Knoll [21], a spectral feature is adequately sampled when its full width at half maximum (FWHM) spans at least five channels. Of the reference sources available for this work, ^{241}Am is the only one that produces a resolvable photopeak in BC-408, at 59.5 keV. The reported energy resolution of BC-408 at this energy is 30 to 50% [80–82]. The narrowest case, 30%, corresponds to a FWHM of 17.9 keV and sets the more demanding limit. Five channels across this FWHM give a channel width of 3.6 keV, and therefore a minimum of 28 channels up to 100 keV. This is a lower bound on the channel count, below which the photopeak is undersampled and energy information is lost. However, it is worth noting that such a small number of channels would be impractical for spectral characterisation, and therefore for energy calibration.

The ^{241}Am source was nevertheless used to guide the gain. The pulses generated by the source were observed directly on the oscilloscope, and the gain was adjusted so that they did not saturate the OpAmps. It was set to remain within an amplitude range that the ADC samples with a reasonable number of channels, leaving room for the higher energies

up to 100 keV. Ultimately, the gain was chosen to give finer binning while still satisfying the condition imposed by the most limiting factor, the resolution of BC-408. The resulting component values of the pulse shaping chain are listed in Table 3.

Table 3. Component values of the pulse shaping chain.

CSA		PZC		Shaper	
Parameter	Value	Parameter	Value	Parameter	Value
R_f	4.7 k Ω	R_{diff}	4.7 k Ω	R_s	10 k Ω
		R_{pz}	4.7 k Ω		
C_f	100 pF	C_{diff}	100 pF	C_s	47 pF
τ	470 ns		470 ns		470 ns

Transient Response

The transient response of the shaping chain to a single ^{241}Am event is shown in Fig. 11. The associated time metrics, extracted over the full set of acquired waveforms, are summarised in Table 4.

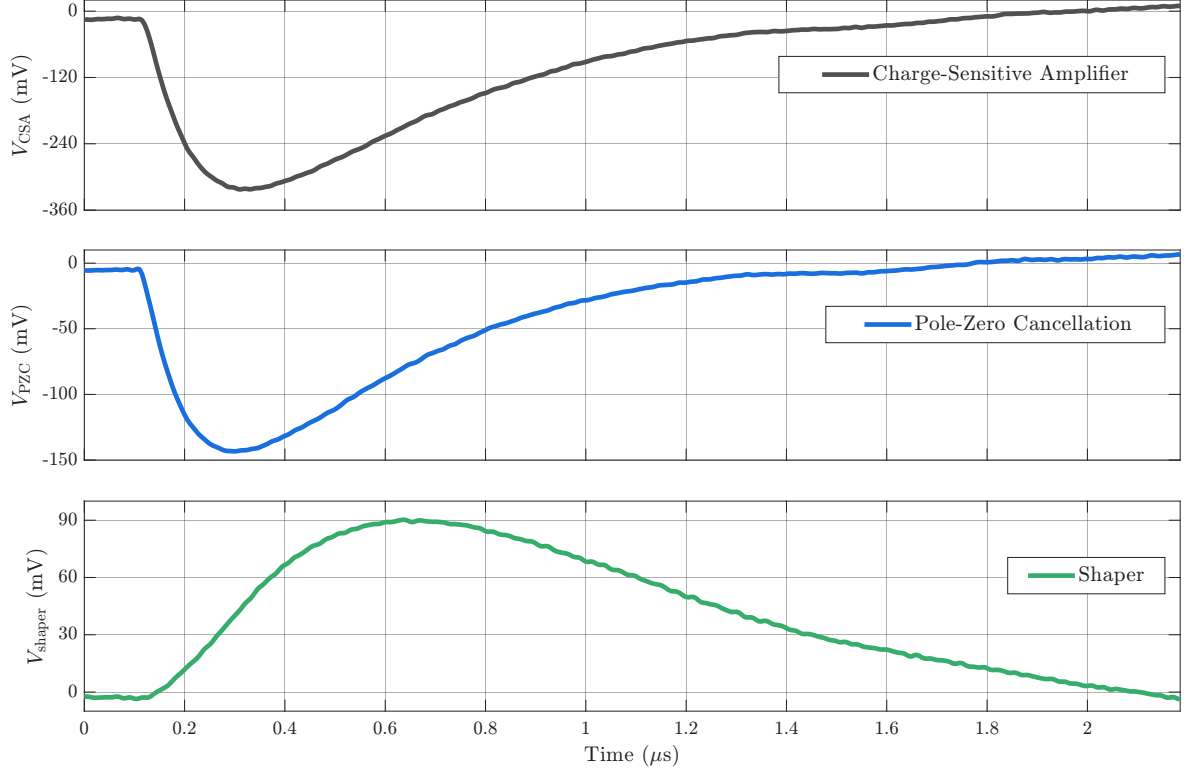


Figure 11: Transient response of the CSA (grey), PZC (blue) and shaper (green) to a single ^{241}Am event.

The integration at the CSA produces a negative pulse, since the charge is collected onto C_f in an inverting configuration, and this polarity is preserved through the differentiating network. The inverting topology adopted for the last integrating stage restores a positive output at the shaper, as required by the single-positive-supply operation of the subsequent stages. A residual undershoot is nevertheless observed at every stage, originating from the CSA. Its relative magnitude is not increased along the chain, which would be the case in the absence of R_{pz} . The gain at the final stage was decreased, so that events of this energy are sampled towards the lower end of the available bins, as discussed.

Table 4. Timing metrics of the pulse shaping chain at each stage.

	CSA	PZC	Shaper
Amplitude range (mV)	191–861	87–430	61–407
Peaking time (ns)	147 ± 28	135 ± 25	387 ± 50
Width at 90% of peak (ns)	209 ± 11	188 ± 12	397 ± 22
Pulse duration (μ s)	1.42 ± 0.09	1.26 ± 0.07	1.96 ± 0.09

The main effect of the shaping is the redistribution of the signal in time. The semi-Gaussian output is broader and the time taken to reach the maximum is increased from 147 ± 28 ns at the CSA output to 387 ± 50 ns at the shaper, even though the latter has a lower amplitude. The interval over which the signal remains within 90% of its maximum is doubled, from 209 ± 11 ns to 397 ± 22 ns. These effects are intended, since the amplitude is retained through comparator logic, and a maximum that is both delayed and sustained reduces the timing demands imposed on both the hardware and firmware. Within a broad range of amplitudes, the total pulse duration is consistent across events, as it is set by the time constants selected for this network. At the shaper output, the pulse lasts 1.96 ± 0.09 μ s. This is the pulse that the subsequent stages discriminate and sample.

2.4 Pulse Discrimination

Pulse discrimination distinguishes real X-ray events from electronic noise and retains the amplitude of those accepted for digitisation. This process is coordinated by the MCU firmware, summarised in Fig. 12 and detailed in the following subsections. In general terms, after the chain is initialised (grey), the MCU waits at an event decision (red), driven by the comparator output (green). Each accepted event then starts a readout sequence (yellow), during which no further triggers are accepted.

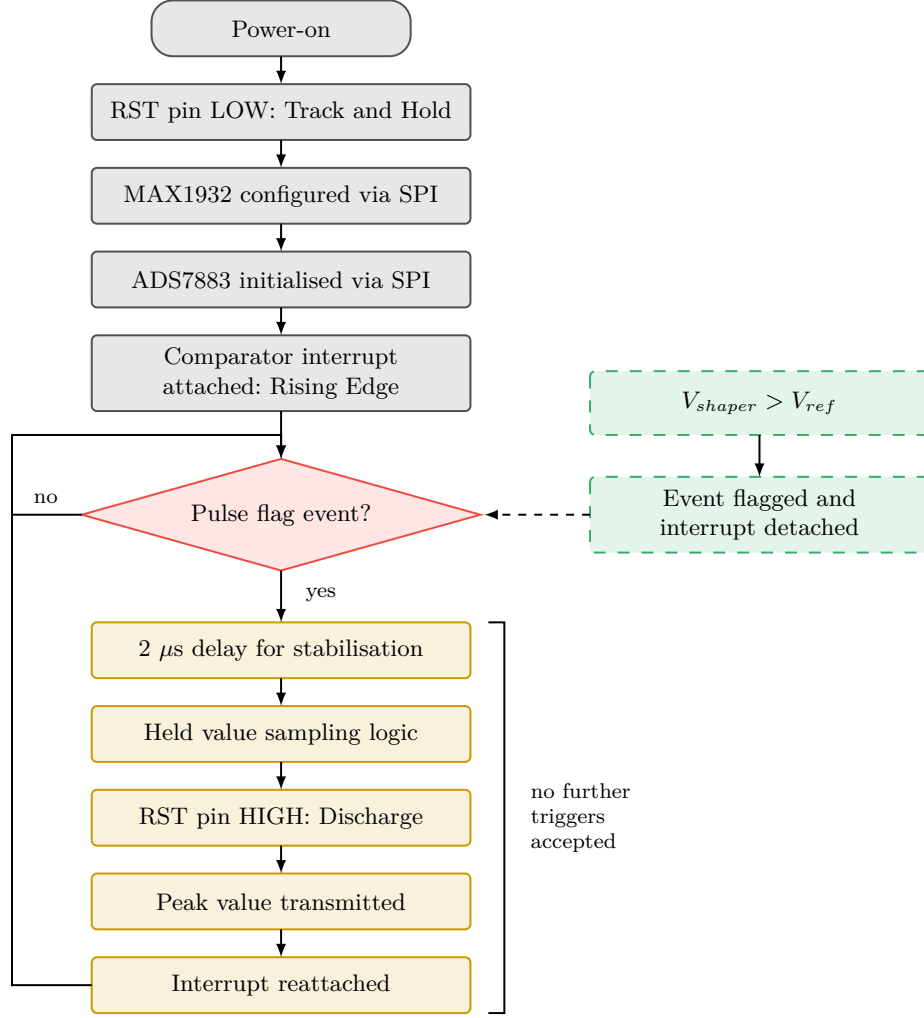


Figure 12: Firmware workflow diagram.

2.4.1 Comparator

As mentioned, discrimination operates on a comparator logic. The LT1711 [83] was selected for this application, a fast comparator operated in a non-inverting configuration. As shown in Fig. 9, the signal from the shaper is applied to the non-inverting input, while a fixed reference voltage, V_{ref} , is applied to the inverting input. The reference is derived from the 3.3 V supply rail through a resistive divider:

$$V_{ref} = \frac{R_2}{R_1 + R_2} \cdot 3.3, \text{ unit: [V]} \quad (2.8)$$

where R_1 is the resistor connecting the node to the supply, and R_2 the resistor between the divider node and ground. The threshold was set to 10 mV. This value was established from the shaper output acquired under dark count conditions of the SiPM. The amplitudes of these events lie predominantly below 10 mV. The condition is satisfied by $R_1 = 33 \text{ k}\Omega$ and $R_2 = 100 \Omega$.

A dedicated pin of the ESP32-C3 is connected to the logic output of the comparator and

configured as an interrupt. Whenever the input exceeds V_{ref} , the comparator output switches to the logic HIGH level. The pin is monitored continuously the dedicated hardware within the MCU, so the transition is registered as soon as it occurs. Detection initiates the readout sequence of Fig. 12, and the interrupt is detached for its duration. It is reattached once the loop has terminated.

2.4.2 Peak-and-Hold

Simultaneously, the shaper output is fed to a Peak-and-Hold (P&H) circuit. A P&H captures the maximum of the input signal and sustains it after the pulse has decayed, extending the interval over which the peak amplitude remains available. The analog conversion can then be performed more accurately. Given the small amplitude of the events, a configuration based on a precision rectifier was adopted [84]. The circuit alternates between two modes. While the shaped pulse is rising, the amplifier is enclosed in a feedback loop that charges a hold capacitor C_H , and the capacitor voltage follows the input. This is the track mode. Once the maximum has passed the loop is broken and the peak voltage is retained, read through a buffer. Both amplifiers are implemented with the OPA354 [85], whose input draws negligible current, so the stored charge is not drained by the buffer. This is the hold mode. The value $C_H = 470$ pF reflects a compromise between the two. A larger capacitance reduces droop during hold, but is charged more slowly, and the higher pulses would no longer be tracked to their maximum [84,86].

With no reset path, the capacitor would only ever accumulate charge. Once a maximum has been stored, a subsequent event of lower amplitude cannot forward bias the charging path. The output would therefore converge to the largest amplitude ever received rather than to the amplitude of each individual event. A discharge path is therefore required, and it must be opened and closed under control so that C_H is emptied between events. This is the function of the switch represented in Fig. 9, implemented in this configuration as an N-channel MOSFET placed in series with $R_H = 7.5$ k Ω across the capacitor, driven by a logic level from the MCU. When held LOW, the path between drain and source is open and the stored charge is retained. At the 3.3 V level the channel is established and the capacitor discharges through R_H with a time constant $\tau = R_H C_H$. The gate is therefore held LOW by default, so that the circuit is always ready to capture the maximum of the signal. Detection of a valid pulse raises the interrupt described above, after which a delay of 2 μ s is imposed to allow the held value to stabilise before the sampling logic is applied. The gate is then driven HIGH for an interval longer than τ , sufficient for C_H to discharge, and returned LOW [84].

Transient Response

The output of the pulse discrimination chain to a single ^{241}Am event is shown in Fig. 13. The trigger level was taken as the intersection between the shaper output and the comparator transition. It is measured at 9.3 mV, in agreement with the nominal threshold of approximately 10 mV. Upon reaching the maximum, the P&H circuit retains the amplitude of the event, although a droop of the held value is present throughout the hold interval. The decay is more pronounced over the first two microseconds after the peak, which justifies the pre-sampling delay mentioned above. The residual droop over the rest of the hold is attributed to leakage through the capacitor, the diode and/or the reset path [84,86].

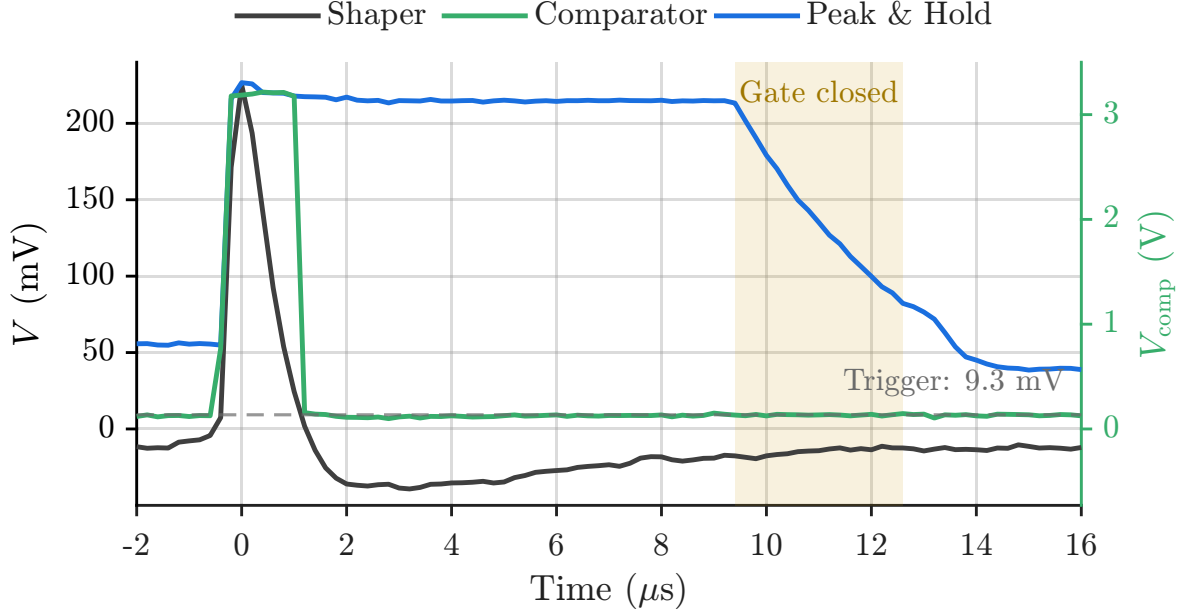


Figure 13: Transient response of the shaper (grey), comparator (green) and P&H (blue) stages to a single ^{241}Am event. The yellow region marks the interval during which the reset transistor is driven HIGH, allowing the hold capacitor to discharge.

No sampling logic was implemented for these measurements. Nevertheless, a latency is present in the transition of the transistor logical state. The hold interval is therefore extended beyond the value imposed by the firmware, which contributes directly to the dead time of the detector. Once the reset path is closed, the discharge of the storage capacitor is governed by the time constant imposed by R_H and C_H . It was observed, however, that the output does not return to ground, settling instead at a residual baseline of approximately 50 mV. The discharge also continues after the gate has returned to the LOW logical state, at which point the capacitor should retain its stored voltage. The latter is consistent with the transition latency described above. It was first considered whether R_H was too high for the discharge to complete within the time the gate sits at 3.3 V. To assess this, R_H was changed to 1 k Ω . Although the discharge was completed in a shorter interval, as expected, the output would settle at the same residual baseline. This behaviour could be explained by the single-supply configuration of the OpAmps. With their negative rail connected directly to ground, the amplifier cannot reproduce the negative baseline of the shaper output. Once the reset path is open, and the input of the amplifier remains negative, the current needed to supply this load may cause an output swing. This output swing would be the floor at which C_H can be minimally charged. Therefore, no event below 50 mV can be sampled, as it cannot be tracked. It is a limitation of the current front-end electronics, and it reduces the dynamic range available at the lower end of the energies of interest. Ultimately, R_H was kept at 7.5 k Ω . Without sampling logic, the mean dead time is measured as $13 \pm 1 \mu\text{s}$, taken as the interval over which the P&H output remains at hold until it discharges to the residual baseline.

2.4.3 Pulse Sampling

Pulse sampling digitises the held signal through an ADC. The converter maps the input range onto 2^n discrete levels, where n is its resolution in bits. Each amplitude is sampled and quantised to a corresponding digital bin. Since the conversion occupies the acquisition chain, the introduction of a sampling stage is expected to increase the dead time of the detector. Two converters were therefore evaluated, the internal ADC of the ESP32-C3 and an external ADS7883 [87], both of 12-bit resolution. The measured dead times are presented in Table 5.

Table 5. Dead time of the acquisition chain for each sampling condition. Values are the mean of multiple ^{241}Am events with the corresponding standard deviation.

Sampling Condition	Dead Time (μs)	Sampling Protocol
<i>N/A</i>	13.2 ± 0.9	<i>N/A</i>
Internal MCU ADC	59.2 ± 1.5	Analog Read
External ADC	17.8 ± 0.4	SPI

From the perspective of power consumption, component complexity and cost reduction, the internal converter would be preferable. However, relative to the dead time measured without a sampling stage, the internal converter adds a delay ten times higher than the external one. The internal ADC is accessed through the `analogRead` routine, and limited by its own execution time, making it unsuitable for this application. On the other hand, the ADS7883 is read over SPI at 20 MHz, a dedicated high-speed protocol, and consequently increasing the dead time by only $4.6 \mu\text{s}$. The external converter was therefore adopted, as it allows a higher count rate to be achieved. On the ESP32, each 12-bit value is reduced to 10 bits by a two-bit right shift before being transmitted. Over the 3.3 V input range, the resulting 1024 channels correspond to a channel width of 3.2 mV, which sets the smallest amplitude difference that can be resolved. With a dead time of $17.8 \pm 0.4 \mu\text{s}$, the maximum count rate of the detector is $(56 \pm 1) \times 10^3$ counts per second.

2.5 Validation of the Front-End Electronics

Prior to energy calibration of the PS, the readout electronics were validated using a $5 \times 5 \times 10 \text{ mm}^3$ CsI(Tl) crystal. The linearity of the energy-to-channel mapping can be tested by relating pulses of known energy to the channels in which they are stored. Unlike the continuous spectra of the X-ray tube and of scattered radiation, a radionuclide emits γ -rays at discrete energies, which provide these known references. To identify them as peaks, photoelectric absorption must be favoured. An inorganic scintillator is appropriate for this validation owing to its higher density and effective atomic number, which favour photoelectric absorption in the energy range of interest. Its higher light yield also gives a much better energy resolution than that of organic scintillators, so the peaks are narrow and their positions well defined. A linear channel-to-energy mapping is expected, and any significant deviation would be attributable to the front-end electronics [21]. This validation therefore indicates whether the front-end electronics introduce any non-linearity, such as the amplitude droop in the P&H or its residual baseline. Should either prove amplitude-dependent, it would require characterisation or correction.

The hardware gain was adjusted to accommodate the higher light output of CsI(Tl). Spectra were then acquired over 10-minute intervals using the three radioactive sources described previously, and a background spectrum was acquired without any source over the same interval. The resulting pulse height spectra are shown in Fig. 14, alongside a simplified decay scheme of each source, restricted to its most probable transitions. The background was subtracted, and channels 18 and below were excluded, as their elevated count rate would otherwise dominate the vertical scale. Given the ADC step, channel 18 corresponds to approximately 58 mV, which lies near the residual baseline of the P&H stage. Events that exceed the 10 mV comparator threshold but remain below this level leave the P&H output at its baseline, and are therefore all assigned to channel 18. These include real events, as they lie above a threshold intended to exclude dark counts from the SiPM. Their energy, however, is masked. This does not compromise the validation, since all expected spectral features are resolved for each source, including the lowest energy present, the 14.4 keV γ -ray emission of ^{57}Co . Since the baseline is a property of the P&H rather than of the gain, the implication for the PS calibration is that the proposed energy range must lie above channel 18.

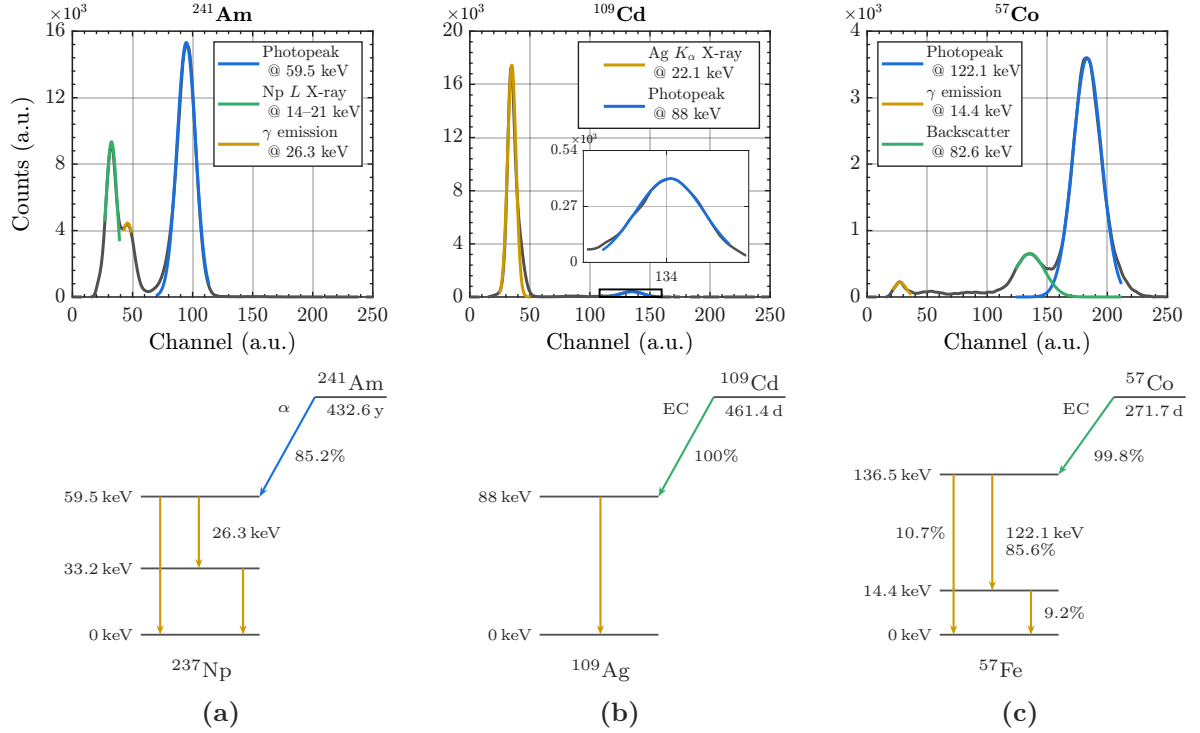


Figure 14: Pulse height spectra acquired with the CsI(Tl) crystal for the three calibration sources, (a) ^{241}Am , (b) ^{109}Cd and (c) ^{57}Co , with the decay scheme of each source beneath.

^{241}Am decays by α emission, predominantly to the 59.5 keV level of ^{237}Np , whose de-excitation to the ground state produces the photopeak at that energy. A smaller fraction of that level instead de-excites through the intermediate 33.2 keV state, and the emission observed at 26.3 keV corresponds to the difference between the two [88]. The remaining two sources decay by electron capture. ^{109}Cd populates the 88 keV excited state of ^{109}Ag , and the peak at 22.1 keV is the K_{α} X-ray of silver. ^{57}Co decays to the 136.5 keV level of ^{57}Fe ,

which de-excites predominantly to the 14.4 keV level and gives the photopeak at 122.1 keV. The direct transition to the ground state at 136.5 keV is not resolved due to the detector's own resolution. The emission observed at 14.4 keV is the subsequent de-excitation of that intermediate level [88]. Finally, the peak at 82.6 keV is the backscatter peak of the 122.1 keV emission, formed by photons scattered at 180° in the material surrounding the crystal and subsequently absorbed within it, as described in section 1.2.2. The centroid channel of each photopeak was mapped to its known energy value and the 22.1 keV line of ^{109}Cd was included, since it adds a reference point at the lower end of the range of interest. A linear fit of the form:

$$C = aE + b, \quad (2.9)$$

was applied, where C is the channel number and E is the energy in keV (Fig. 15). The fit yields $R^2 = 0.999$, showing a linear channel-to-energy relationship up to at least channel 183, corresponding to an input amplitude of approximately 590 mV. This confirms that the readout chain does not introduce amplitude-dependent errors within this bin range.

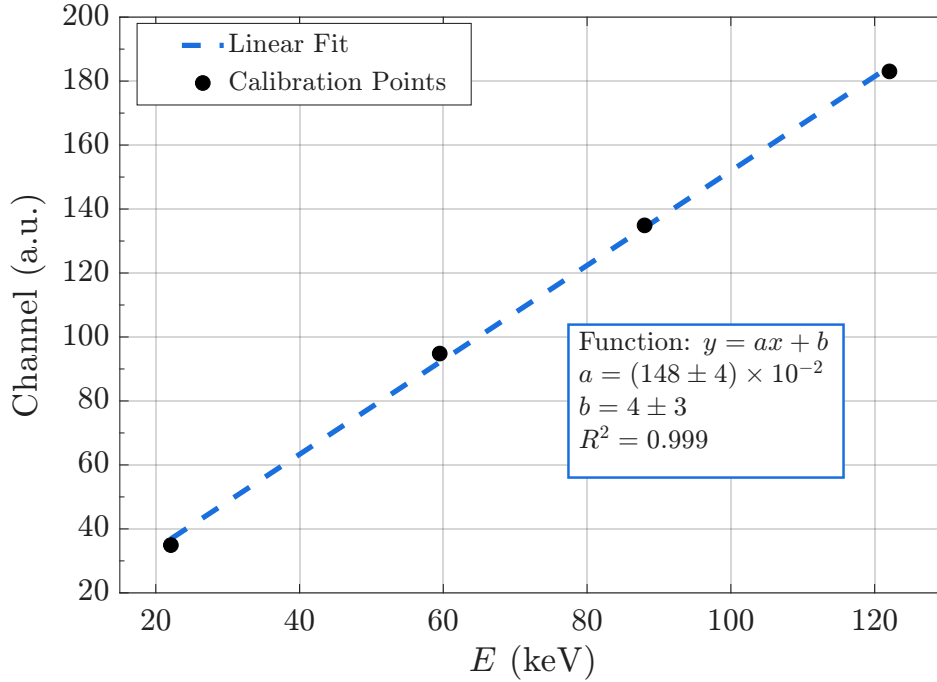


Figure 15: Channel-to-energy calibration of the front-end electronics obtained with the CsI(Tl) crystal, with a linear fit.

By reconstructing the ^{57}Co spectrum in energy units, the 122 keV photopeak shows a FWHM of 19 ± 1 keV, corresponding to an energy resolution of $16 \pm 1\%$ at 122 keV, consistent with values reported in the literature for CsI(Tl) [89, 90]. This indicates that the front-end electronics do not significantly degrade the intrinsic energy resolution of the crystal. The front-end is therefore considered validated.

Calibration of the Prototype

3.1 Energy Calibration

As an approach for PSs, the energy response was calibrated using the *Bremsstrahlung* endpoint of the X-ray tube available. The maximum X-ray photon energy produced corresponds to the scenario in which the accelerated electron transfers all its kinetic energy in a single interaction with the anode nucleus. The maximum energy detected, i.e., the endpoint of the acquired spectrum, is therefore equal to the tube voltage, which serves as the energy reference for calibration. Since the *Bremsstrahlung* spectrum near the endpoint is approximately linear, this abscissa can be determined as the x -intercept of a least-squares fit to that region [91].

The detector was positioned in the beam line, at a distance sufficient to avoid saturation of the front-end electronics. *Bremsstrahlung* spectra were acquired over 5-minute intervals for tube voltages of 15–50 kVp and tube currents of 0.050–0.250 mA, and are shown in Fig. 16 (a) and (b), respectively. Background was also subtracted, and channels 18 and below were again excluded. As represented in Fig. 16 (a), the 15 kVp spectrum is not visible, as the signal produced at the tube voltage and any current is low enough to be detected. This agrees with the preliminary measurements of Section 2.1.2, in which 15 kVp produced the signal closest to the dark current. At this distance and fluence, photons of such low energy may be absorbed by the black adhesive tape used for light-wrapping of the photosensitive detector element. This establishes the lower bound of the calibrated range at 20 keV.

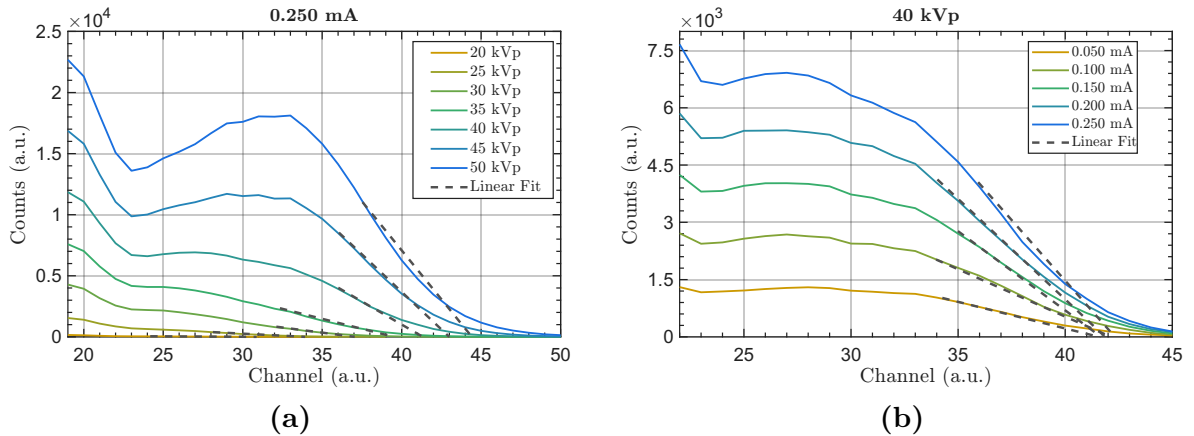


Figure 16: *Bremsstrahlung* spectra, background subtracted and excluded; (a) acquired at 0.250 mA for tube voltages from 15 to 50 kVp; (b) acquired at 40 kVp for tube currents from 0.050 to 0.250 mA.

The linear region of each *Bremsstrahlung* spectrum was selected manually, by visual inspection, and a least-squares regression was applied to the points within that window. The

x -intercept of the fitted line gives the endpoint channel. Since tube current affects only photon fluence and not the spectral endpoint, the five acquisitions per voltage produce five independent estimates (see Fig. 16 (b)). The resulting values are summarized in Table 6.

Table 6. *Bremsstrahlung* endpoint channel for each tube voltage. Values are the mean of five acquisitions with the maximum deviation from the mean.

Tube Voltage (kVp)	20	25	30	35	40	45	50
Endpoint Channel (a.u.)	31.0 ± 1.0	33.4 ± 0.9	36.6 ± 0.4	39.3 ± 0.3	41.9 ± 0.4	43.3 ± 0.5	45.1 ± 0.8

Although 20 and 25 keV lie at the boundary at which photoelectric absorption ceases to dominate in BC-408 (see Appendix B), the associated uncertainty is dominated by counting statistics. The associated error improves between 30 and 45 keV, indicating that in this region both the photoelectric cross-section and the photon fluence are sufficient for the endpoint to be determined with good precision. At 50 keV, the photoelectric absorption in the scintillator drops below 0.5%, and consequently the uncertainty increases. As full energy transfer becomes less probable, the spectrum compresses toward a lower and less reliable endpoint. Additionally, pileup further distorts the spectral shape near the abscissa. The pileup distortion is visible as a non-linear tail in the higher energy spectra of Fig. 16 (a), which is not characteristic of filtered *Bremsstrahlung*. At higher count rates, the baseline does not recover between successive pulses, causing them to superimpose and producing the observed tail. These are the fundamental limitations of the endpoint method at higher energies, as the associated uncertainty is expected to grow accordingly.

To extend the calibration beyond the range of the X-ray tube, spectra of the reference sources were acquired with the PS five times, each over a 90 min interval. Of the three sources, the only features available are the ^{241}Am photopeak at 59.5 keV, and the Compton Edge (CE) of the 122 keV line of ^{57}Co , at 39.46 keV, shown in Fig. 17 (a) and (b) respectively.

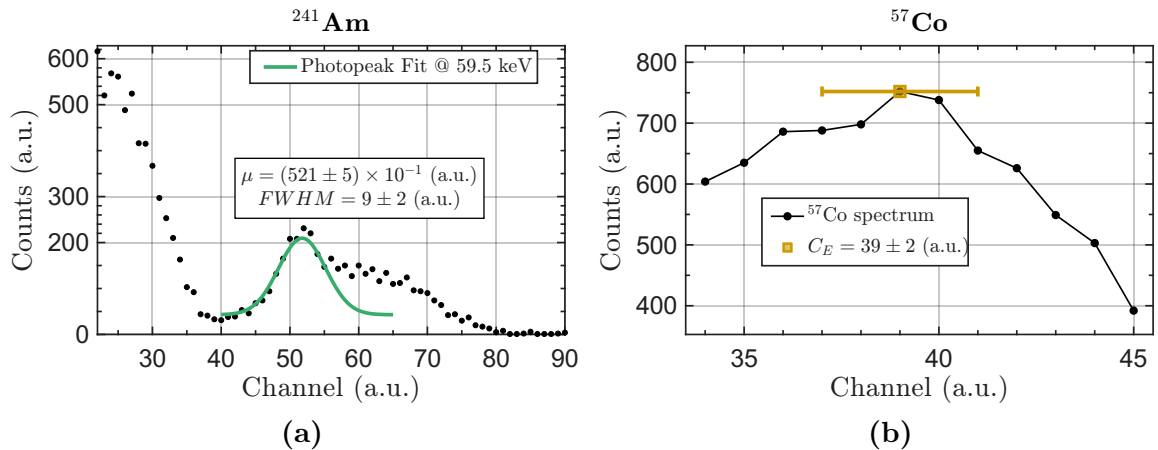


Figure 17: Reference source spectra, background subtracted. Annotated values are the mean of five acquisitions with the maximum deviation from the mean as uncertainty. (a) ^{241}Am , with the Gaussian fit to the 59.5 keV photopeak. (b) ^{57}Co 's Compton Edge at 39.46 keV.

The ^{241}Am photopeak is fully resolved and was described by a Gaussian fit, whereas the ^{57}Co CE was extracted as the channel of maximum counts. Despite the lower number of counts accumulated in the photopeak, it has the smaller uncertainty of the two features, and although it is not as sharp as the one obtained with the CsI crystal in Section 2.5, the centroid position is reproducible across acquisitions. The broadening of the ^{57}Co feature is a consequence of the energy dependence of the resolution and its degradation at lower energies. The photopeak is therefore included as a calibration point at 59.5 keV, whereas the edge, falling within the interval already covered by the tube, is treated as an independent validation of the *Bremsstrahlung* method.

A weighted least-squares fit was applied to account for the differences in uncertainty across the energy range and is shown in Fig. 18.

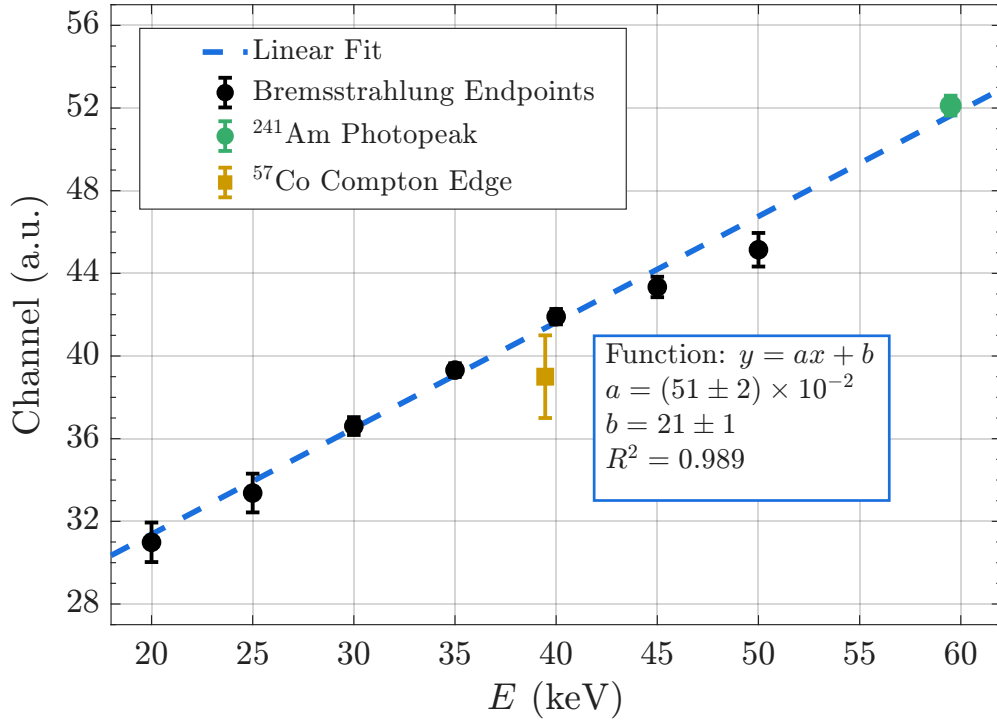


Figure 18: Weighted least-squares energy calibration of the prototype, obtained from the seven *Bremsstrahlung* endpoints and the ^{241}Am photopeak centroid. The ^{57}Co Compton edge is superimposed for validation.

Although the 45 keV uncertainty is small, both the 45 and 50 keV endpoints fall below the calibration line, beyond their uncertainties. This underestimation is consistent with the compression of the spectrum discussed above. In contrast, the ^{241}Am centroid falls on the calibration line with an uncertainty smaller than that of the highest-energy endpoint estimate. The weighted fit ensures that these better-determined points contribute more to the calibration, and the uncertainties in each estimate propagate into the calibration coefficients [91]. The fit is of the form:

$$C = aE + b, \quad (3.1)$$

where C is the channel number and E is the energy in keV. With $R^2 = 0.989$, the response can be treated as locally linear within the calibrated interval of 20 to 60 keV. Given a measured

channel, the energy is therefore obtained from the following simplified expression [92]:

$$E = \frac{C - 21}{0.51}, \quad (3.2)$$

$$\sigma_E = \frac{1}{0.51} \sqrt{(0.02)^2 E^2 + 1}. \quad (3.3)$$

The ^{241}Am photopeak in Fig. 17 (a) has a FWHM of 9 ± 2 channels, above the minimum of five channels set in Section 2.3. The chosen gain therefore samples the photopeak adequately. As the only available photopeak, its centroid is also used to calculate the energy resolution of the system at 59.5 keV. Although R is dimensionless and, in principle, independent of the channel-to-energy conversion, the non-zero intercept b shifts the centroid without contributing to the FWHM. Consequently, the resolution differs between channel and energy space and these parameters must be converted to energy using Eqs. 3.2 and 3.3. With these values, the dosimeter has an energy resolution of $R = (29 \pm 5)\%$ at 59.5 keV, consistent with values reported in the literature, and previously discussed, for BC-408 PSs.

As for the CE of the 122 keV line of ^{57}Co , the extracted channel falls below the calibration line, and remains outside it even when its own uncertainty is considered. This discrepancy is attributed to the extraction method rather than to the calibration itself. According to Swiderski *et al.* [82], the finite resolution of PSs causes the channel of maximum counts not to coincide with the true position of the edge. The CE lies at a channel above that of maximum counts, and the offset grows as the resolution degrades. Taking the CE as the maximum of the distribution therefore introduces a systematic underestimation, as observed here. At these energies, the CE is thus not a reliable calibration point. At higher energies, where the resolution improves and the CE is routinely used to calibrate PSs [82], it may provide a way of extending the calibration beyond the present range.

Without such points, extension of the linear fit to 100 keV is assumed, as it would span the non-proportional region of the plastic response discussed in Section 1.5.2. The calibration is therefore restricted to the points obtained with the available sources, and the gap between 60 and 100 keV is acknowledged as a limitation of the present work. Within these limits, the prototype response is characterised across the 20 to 60 keV range, and the following dose calibration is performed within this interval.

3.2 Dose Calibration

As introduced, an individual monitoring device is calibrated in personal dose equivalent, $H_p(d)$, with $d = 10$ mm adopted, the depth at which the quantity is defined for whole-body exposure. These devices are calibrated in certified laboratories, which have reference instruments and ensure the reproducibility and traceability of the measurements. The available reference fields include radionuclide sources, mono-energetic X-radiation and continuous filtered X-ray beams of defined quality, and conversion coefficients are provided for all of them. The calibration presented here follows the ISO 4037 series [25, 26, 93] for the conversion of *air KERMA* to $H_p(10)$, using filtered X-ray beams of the narrow (N) series, the same reference qualities over which the commercial RaySafe i3 is calibrated. The protocol was adapted to the instrumentation available with the intention of demonstrating the methodology and showing that the proposed dosimeter can be calibrated as a prototype.

3.2.1 Setup

The production of the radiation quality needed depends on the tube potential, the total filtration and the properties of the anode. Given the X-ray tube available and the energy calibration, only radiation qualities with energies in the 20 to 50 keV range can be used. In the N-series that corresponds to N-25, N-30 and N-40, as the N-20 spectrum falls outside the calibrated energy range. The parameters defining each of the three qualities, together with the corresponding conversion coefficients, are listed in Table 7.

Table 7. N-series radiation qualities. Coefficients correspond to the ISO slab phantom at 1-meter normal incidence. Adapted from Refs. [25,26].

Radiation Quality	Tube Potential (kVp)	Added Filtration	$h_{pK}(10)$ (Sv/Gy)
N-25	25	2 mm Al	0.574
N-30	30	4 mm Al	0.815
N-40	40	4 mm Al + 0.21 mm Cu	1.210

As introduced, the dosimeter must be placed on a PMMA phantom during the calibration. For $H_p(10)$ the ISO water slab phantom is used, since it reproduces the backscatter of the trunk of the body and therefore the conditions under which the dosimeter is worn for this specific quantity. It consists of a PMMA shell of $30 \times 30 \times 15$ cm, with a 2.5 mm front wall and 10 mm remaining walls, filled with water [26]. The setup assembled for this work is shown in Fig. 19. A phantom replicating these dimensions was built, but it was not filled with water, due to the risk of leakage in the presence of electronics around the setup. The test board was taped to the front wall of the phantom, with its photosensitive area aligned with the centre of the tube. Custom 3D-printed supports were made to hold the filters, and these sit on an alignment rail together with the phantom. The centre of the PS entrance face is placed 1 m away from the tube, and this position is taken as the reference measurement point, marked by the red star in the figure. The *air KERMA* was measured with the R/F Low detector of the RaySafe Xi [94], which was the only one to provide a measurable signal at this distance. The instrument was last calibrated in October 2013 and the product has since been discontinued, but it was the only reference available. For these measurements the phantom and the dosimeter are removed from the rail and the centre of the R/F Low detector is placed at the reference point.

It should be noted that the reference conditions under which the conversion coefficients of Table 7 are defined cannot be fully reproduced with the available instrumentation. The phantom was not filled with water, so the backscatter contribution is lower than the one assumed by the standard. The anode is made of molybdenum instead of tungsten, although its characteristic lines lie below the calibrated energy range and therefore do not contribute to the measured spectra. The inherent filtration falls only within the maximum deviation permitted by ISO 4037, and the HVLs of each quality were not tracked. Most importantly, the main limitation is the reference instrument itself. The values obtained are therefore not traceable and the procedure described here does not constitute a formal calibration. Its purpose is to establish and validate a methodology that reproduces the ISO 4037 framework as closely as the available setup allows, so that the same procedure can later be planned and repeated at

a certified facility under the required field conditions.

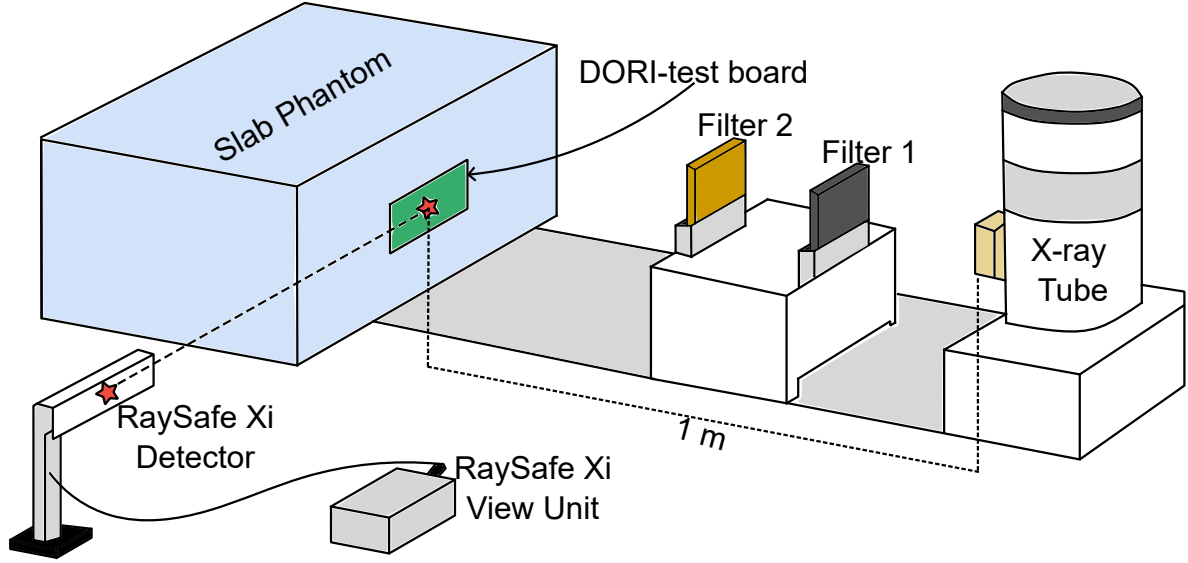


Figure 19: Dose calibration setup. The beam quality is defined by the placement of one filter, or two in the case of N-40, at the exit window of the X-ray tube. The test board is placed at the reference measurement point (red star), in contact with the front wall of the built slab phantom. The RaySafe Xi detector is positioned at the same test point, with the phantom removed, and the *air* KERMA values are read on its view unit.

All acquisitions were made at a tube current of 0.5 mA for 2 minutes, and the spectrum and average *air* KERMA rate from prototype and the RaySafe were recorded, respectively. The spectra acquired are shown in Fig. 20 (a). Counts below the lower limit of the validated energy calibration and above the endpoint energy of the respective beam were excluded. Each spectrum is therefore confined to its expected energy range, and noise and pile-up events are rejected.

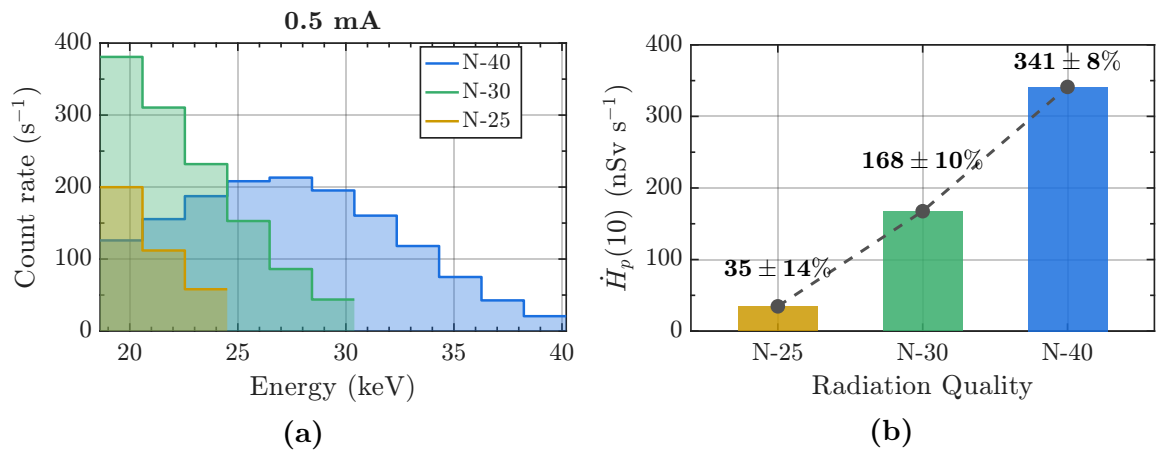


Figure 20: ISO 4037 narrow-series qualities N-25, N-30 and N-40, acquired at 0.500 mA for 2 minutes. (a) Energy spectra; (b) reference $\dot{H}_p(10)$.

The effect of the filtration can be seen, since all acquisitions were made at the same tube current and for the same duration. Without filtration, the count rate would scale approximately with the square of the tube potential. The N-series filtration is designed to narrow the spectrum by suppressing its low-energy component. The lower count rate registered at the lower energies for N-40 is a direct consequence of the copper filter. This is not seen for N-30. N-30 requires twice the aluminium of N-25 and yet exceeds it at every energy the two share, since the higher fluence compensates the additional attenuation. The low-energy component of N-40 is nevertheless not completely suppressed. The detected counts originate from Compton interactions in the PS and from photons scattered by the slab phantom, which is itself intended to produce scattering. As a consequence, the mean energy of each distribution falls progressively below the value tabulated by ISO for the corresponding quality.

The reference $\dot{H}_p(10)$ obtained for each quality is shown in Fig. 20(b), calculated from the measured *air KERMA* rate and the conversion coefficients of Table 7. The values increase with tube potential, as both terms of the product grow with beam energy. The relative uncertainty decreases with tube potential, from 14% at N-25 to 8% at N-40. The *air KERMA* rates of N-30 and N-40 sit above the trigger level of the detector ($> 100 \text{ nGy s}^{-1}$) [95], while that of N-25 is flagged as a low signal, contributing to the higher uncertainty.

3.2.2 Methods

The absorbed dose increases with the energy deposited, so counts registered at different energies do not produce the same biological effect. $H_p(10)$ can therefore be written as a sum over the spectrum, each count contributing according to the energy at which it was detected:

$$H_p(10) = \sum_{i=1}^m w_i N_i, \quad (3.4)$$

where N_i is the number of counts and w_i the dose per count, in Sv/count. Each weight holds the response of the prototype dosimeter to the ISO reference field in that energy range, since it converts the counts the device registers into the dose delivered. Calibrating the dosimeter in dose therefore corresponds to determining the set of weights w_i .

As shown, each radiation quality produces one pulse height spectrum with an associated $H_p(10)$ rate. Since the radiation is not monoenergetic, the weights cannot be assigned to individual energies. The spectrum must instead be grouped into m bins, each covering a range of energies which will carry the same weight. The number of bins that can be resolved is limited by the number of radiation qualities available, as each quality provides one equation of the form of Eq. 3.4. For q radiation qualities, the measurements can be written as:

$$\begin{bmatrix} H_1 \\ H_2 \\ \vdots \\ H_q \end{bmatrix} = \begin{bmatrix} N_{11} & N_{12} & \cdots & N_{1m} \\ N_{21} & N_{22} & \cdots & N_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ N_{q1} & N_{q2} & \cdots & N_{qm} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix}, \quad (3.5)$$

where each row of $\mathbf{A}$ holds the counts per bin measured for one quality, $\mathbf{H}$ is the measured $H_p(10)$ and $\mathbf{w}$ the unknown weights. For $q > m$ the system is overdetermined and is solved by the least-squares method [96]:

$$\mathbf{w} = \left(\mathbf{A}^\top \mathbf{A} \right)^{-1} \mathbf{A}^\top \mathbf{H}. \quad (3.6)$$

With three radiation qualities available, $q = 3$ and the system of Eq. 3.5 admits at most three unknowns. Solving it with $m = 3$ fits the three measurements exactly, leaving no redundancy with which to verify the solution and making no use of the uncertainties of the reference values. Choosing $m = 2$ leaves one degree of freedom, which allows a weighted solution and leaves a residual against which the quality of the fit can be assessed. For this case, the counts of every quality are summed into two groups separated by a single boundary. Eleven channels cover the 20 to 40 keV range, so ten boundary positions are possible. All of them were tested and the two most representative cases are shown in Fig. 21 (a) and (b), the division at 25 keV and the equal-width division at 30 keV, respectively.

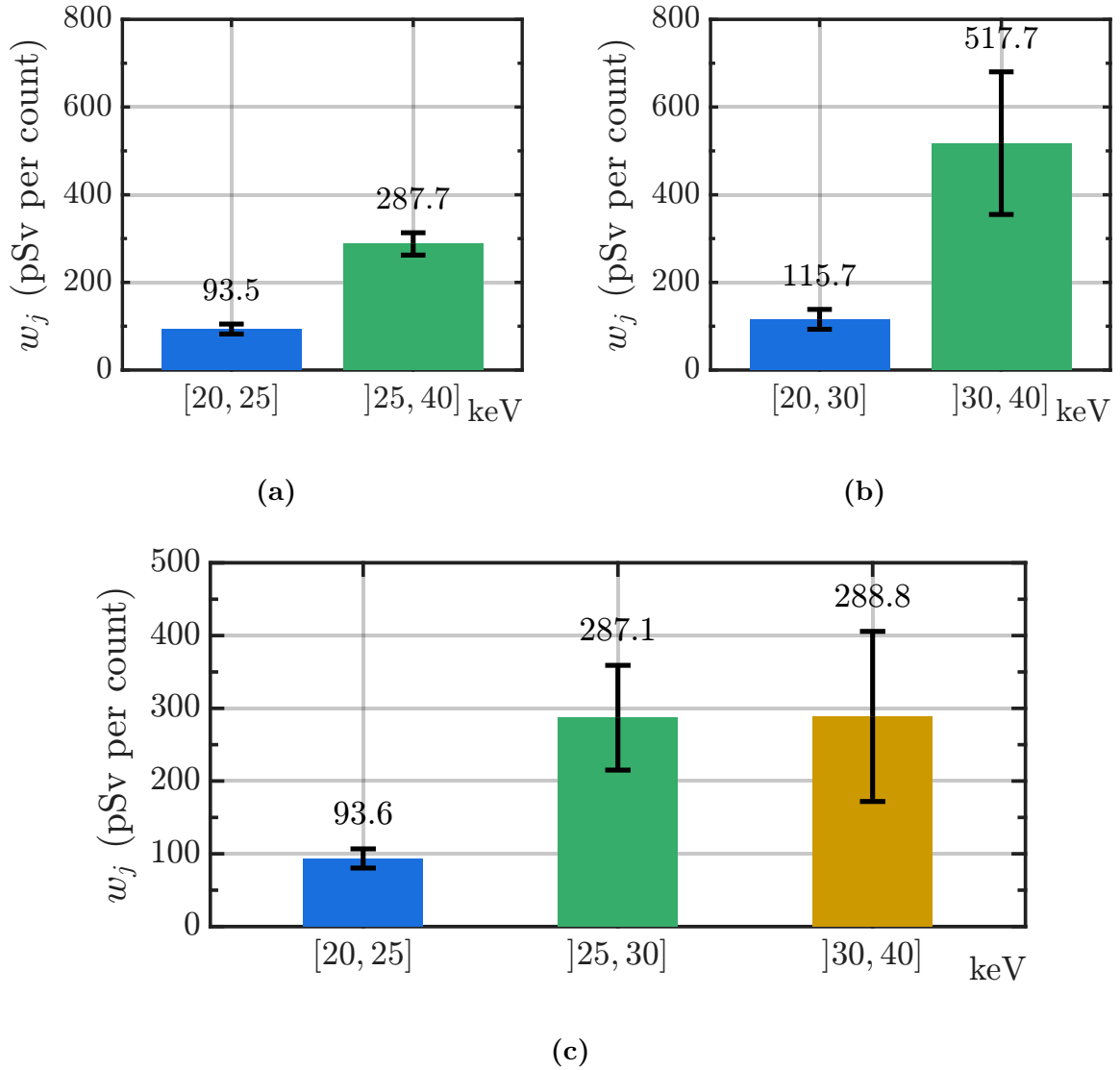


Figure 21: Dose per count obtained for three binning schemes within the calibrated 20 to 40 keV range. (a) Two bins divided at 25 keV; (b) two bins divided at 30 keV; (c) three bins divided at 25 and 30 keV.

Both divisions return $w_2 > w_1$, as expected from the increase of dose with the energy detected, but they differ in how well each weight is determined. At 25 keV the boundary

coincides with the end of the N-25 spectrum. N-25 therefore falls entirely in the lower bin and fixes w_1 by itself, while N-30 and N-40 both populate the upper bin and together determine w_2 . Every quality is used, and the uncertainties obtained are the lowest of the three schemes. For any boundary above this energy, N-25 and N-30 both contribute to the lower bin, and for these cases, most notably at 30 keV, the two qualities have incompatible values of w_1 , 93.6 and 139.0 pSv/count respectively. The solution is therefore a compromise between the two. Additionally, and as represented for this specific scenario, the uncertainty of w_2 is the largest, which makes this division unsuitable.

The small number of measurements also allows the solution to be determined directly. As mentioned in the last paragraph, since N-25 has no counts in the upper bin, it determines w_1 on its own, 93.6 ± 13.2 pSv/count. Subtracting that contribution from N-30 gives $w_2 = 287.1 \pm 72.0$ pSv/count over the 25 to 30 keV range, and subtracting both contributions from N-40 gives $w_3 = 288.8 \pm 116.9$ pSv/count over the 30 to 40 keV range. Two independent radiation qualities therefore return the same weight, within the uncertainty of the fit, and the increase expected for the third weight is marginal and lies only within its uncertainty. This reasoning is the three-bin solution represented in Fig. 21 (c). For this scenario, the uncertainties of the two upper weights increase to 25% and 40% against 10% in the two-bin solution, which makes the model less robust.

A two-bin model was therefore adopted, with $N_1 \in [20, 25]$ keV and $N_2 \in [25, 40]$ keV, and Eq. 3.4 reduces to:

$$H_p(10) = w_1 N_1 + w_2 N_2, \quad (3.7)$$

with the values in Fig. 21 (a):

$$w_1 = 93.5 \pm 11.4, \quad w_2 = 287.7 \pm 25.5, \quad \text{unit: } \left[\frac{\text{pSv}}{\text{count}} \right]. \quad (3.8)$$

A count registered in the upper bin therefore carries 3.1 times the dose of a count registered in the lower bin, which translates the effect of higher energy on the dose delivered. It should be noted that a 15 keV bin is broad considering the available range, but it is again a consequence of the small number of radiation qualities that can be used. A tube capable of higher potentials would provide additional qualities and allow finer divisions. Given a measured spectrum, the personal dose equivalent is therefore obtained from the following simplified expression:

$$H_p(10) = 93.5 N_1 + 287.7 N_2, \quad (3.9)$$

$$\sigma_{H_p(10)} = \sqrt{(11.4 N_1)^2 + (25.5 N_2)^2}, \quad (3.10)$$

The response of the dosimeter after calibration is shown in Table 8, where the reconstructed $H_p(10)$ agrees with the reference value within the uncertainty, for the three qualities and for both reconstructions. The calibration therefore reproduces the measurements from which it was derived. Additionally, a leave-one-out test was performed, where the weights are recomputed from two qualities and used to predict the $\dot{H}_p(10)$ of the third by reconstructing the dose from its own bin counts. The most interesting result is the case of N-25. As mentioned, it is the only direct measurement of w_1 . When excluded, w_1 can only be inferred from the different proportions in which N-30 and N-40 divide their counts between the two bins. As can be seen, the reconstructed dose differs by approximately 0.3%, indicating that w_1 is well determined by the available measurements and is not an artefact of the single quality that defines it.

Overall, these results support the two-bin model and the position of its boundary, and indicate that, with the three measurements available, this was the best division that could be made. The precision of the calibration is limited not only by the number of qualities, but also by the uncertainty of the reference dose values, which is dominated by the RaySafe Xi for the reasons already given. It should be emphasised that a formal calibration was never the purpose. The goal of demonstrating that a prototype of this kind can be calibrated in dose from its own spectrum has been achieved, and the methodology is shown to be workable. A significant part of the future work will therefore be to extend the energy calibration to higher energies and to obtain more reliable $\dot{H}_p(10)$ reference values.

Table 8. Reference $\dot{H}_p(10)$ and the values reconstructed by the two-bin calibration and by the leave-one-out test, for the three radiation qualities.

Radiation Quality	Reference (nSv s ⁻¹)	Calibration (nSv s ⁻¹)	Leave-one-out (nSv s ⁻¹)
N-25	35 ± 5	35 ± 4	35 ± 8
N-30	168 ± 16	168 ± 13	168 ± 14
N-40	341 ± 26	341 ± 27	340 ± 75

3.3 Dose Rate Reconstruction

The spectra acquired for the energy calibration were reprocessed through the two-bin dose calibration. Each spectrum was cut at its own *Bremsstrahlung* endpoint, and only tube potentials of 25, 30, 35 and 40 kVp were used, as they sit within the calibrated energy range. The resulting dose rates, for five tube currents at each of the four potentials, are shown in Fig. 22.

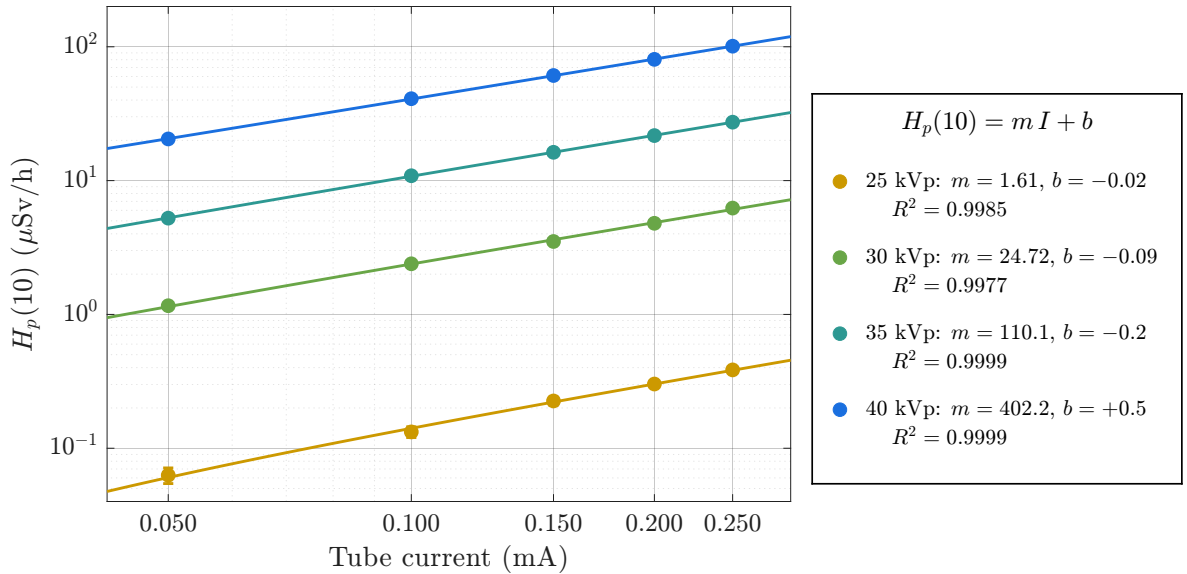


Figure 22: Reconstructed $H_p(10)$ rate as a function of tube current, for tube potentials of 25–40 kVp and currents of 0.050–0.250 mA, on logarithmic axes. The lines are linear fits to the measured values.

The measurements span 0.063 to 101 $\mu\text{Sv/h}$. At every tube potential the dose rate increases linearly with the tube current, with $R^2 \geq 0.998$. The fitted intercepts are consistent with zero at all four potentials, as expected, since no tube current means no counts and therefore no dose. The largest deviation from the fitted line is 6%, occurring at 25 kVp, and is attributed to the poorer counting statistics at this potential. The total count rate over the whole set ranges from 82 to $1.9 \times 10^3 \text{ s}^{-1}$, against the saturation count rate of $56 \times 10^3 \text{ s}^{-1}$ given in Section 2.4.3. The measurements therefore lie within a reliable photon counting region, where dead time and pile-up are negligible. The ratio N_2/N_1 remains constant with the tube current at the potentials where both bins are populated, indicating that the distribution of counts is preserved over this range.

The relevance of this range can be assessed against the isodose map of Fig. 4, keeping in mind that the reliability of the reconstructed values depends on the assumptions of the dose calibration described in the previous section. That map spans 0.0075 to 167 $\mu\text{Sv/h}$ behind a lead apron and 0.5 to $12 \times 10^3 \mu\text{Sv/h}$ without one, at chest level, for one of the most demanding procedures in this modality. The range reconstructed here overlaps most of the shielded scale, but covers only the lower part of the unshielded one, which are detected at higher scattering distances. Other studies report average dose rates of approximately 30 $\mu\text{Sv/h}$ for conventional interventions above personal shielding [97]. The dose rates are therefore relevant, but the position at which the dosimeter is worn should be considered when a full energy and dose calibration is defined. The limit in count rate has also not been established, and the performance of the prototype has to be characterised before an operating range in dose rate can be claimed. Reaching those rates would involve higher photon energies, and establishing the limit of linearity requires, once again, a more complete energy and dose calibration.

Final Prototype

The current prototype is shown in Fig. 23. A new PCB was designed with emphasis on a more compact footprint, correcting the layout errors identified in the test board and optimised for wireless operation. In comparison with the previous PCB, the colour-coded LED was integrated, and a power switch was added to disconnect the LiPo battery from the circuit. Additionally, all oscilloscope test points were removed and the SiPM is now soldered directly onto the board.

To obtain a portable version of the system, and to ensure that no ambient light reaches the photosensitive elements, a two-part 3D printed case was designed. The lower layer contains a dedicated compartment for the LiPo battery (Fig. 23 (a)) and is enclosed by the PCB. As shown in Fig. 23 (b), the PS was wrapped in several layers of Teflon tape, covering all non-exiting surfaces of the cube. The same optical silicone grease was applied as coupling between the SiPM and the PS. The friction it creates, together with dedicated holes in the top layer of the case, keep the PS in place and aligned with the SiPM. Apertures are provided for the LED, the power switch (Fig. 23 (c)) and the USB-C port for charging and firmware uploading. The top layer is closed tightly against the bottom one with four screws. A loop was modelled on the top to hold a standard badge clip, allowing the unit to be worn over clothing. As represented, the current unit measures 6.3 cm in width, 7.2 cm in height and 2.5 cm in thickness, with a total mass of 141 g. For comparison, the RaySafe i3 device measures $4.0 \times 5.8 \times 1.7$ cm and has a mass of only 34 g. Although the dimensions are comparable, this PCB still leaves room for improvement in size, through a better use of the PCB's top and bottom layers for component placement. Additionally, the integration of the ESP32-C3 as a bare chip, rather than the prototyping module, would contribute further to the reduction of the footprint. From a wearable point of view, the mass is currently the most likely cause of discomfort. The enclosure was printed at 100% infill to guarantee that no ambient light reaches the photosensitive elements. This choice increases the mass of the unit, which is higher than desirable for a badge worn over clothing. If light-tightness can be preserved, a reduction can be achieved with a lower infill density, or with different printing methods and materials.

All design files of the final prototype are made available in a public repository (<https://github.com/martamoniz/DORI>), so that the system can be reproduced and further developed. These include the CAD models of the enclosure, the PCB layout and schematic, the firmware, the acquisition software and the energy and dose calibration parameters derived in this work. Future work should also include the implementation of the graphical user interface projected in Chapter 1.5.2, for real time visualisation of the dose rate and storage of the acquired data.

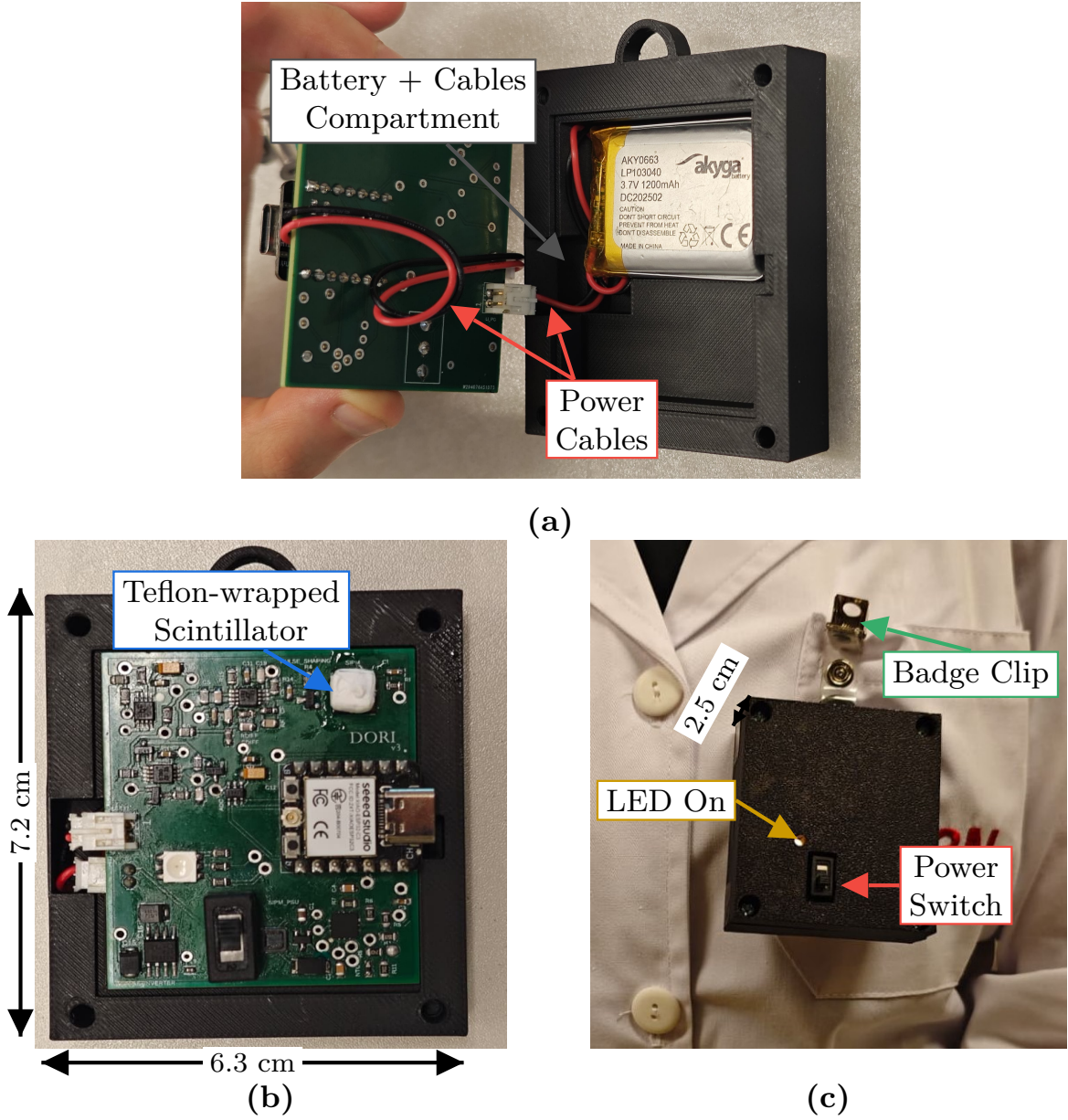


Figure 23: Final dosimeter prototype. (a) Bottom layer of the enclosure, with the compartment holding the LiPo battery and the cables that connect it to the board. (b) Assembled PCB, with the Teflon-wrapped scintillator coupled to the SiPM. (c) Closed prototype worn over a lab coat, with the badge clip, the power switch and the LED on.

4.1 Channel Response

Although the readout circuit is the one described in Section 2.2.2, the response of the new prototype cannot be assumed to be that of the test board. Component values are specified within tolerance bands, and the parasitic contributions and noise coupling introduced by a board are a property of its layout. Both act on the final amplitude at the sampling stage, and consequently on the channel at which a given deposited energy is recorded. Characterising

this response allows the front-end design to be treated as transferable and scalable to different board dimensions and formats. If the response of a new board can be related to that of the fully characterised test board, then the same readout can be used in whatever format each proposed monitoring position demands.

To verify this, the *Bremsstrahlung* endpoint procedure of Section 3.1 was repeated. Spectra were acquired over 5-minute intervals for tube voltages of 20 to 50 kVp, at tube currents of 0.100, 0.150 and 0.200 mA. The endpoint channel was extracted from each acquisition as before. The detector was again positioned within the beam line, although at a distance different from the one used previously. The change was required to prevent saturation of the front-end electronics across the full range of tube voltages, and it is the reason why no spectrum was obtained at 20 kVp. At this distance and fluence the counts registered at that voltage remain within the excluded background. A shorter distance or a higher tube current would recover the point at the cost of saturating the front end at the upper voltages. Additionally, the 3D printed case adds absorption, most significantly at the lower energies. The resulting endpoint channels are summarised in Table 9. The uncertainty on the individual endpoints follows the pattern already described for the energy calibration. At 25 keV it is dominated by counting statistics; it decreases towards the middle of the interval and increases again towards 50 keV, where photoelectric absorption in the scintillator drops and pile-up distorts the spectral shape near the abscissa.

Table 9. *Bremsstrahlung* endpoint channel for each tube voltage, obtained with the final prototype. Values are the mean of three acquisitions with the maximum deviation from the mean.

Tube Voltage (kVp)	25	30	35	40	45	50
Endpoint Channel (a.u.)	$30 \pm 4\%$	$34 \pm 1\%$	$36 \pm 1\%$	$39 \pm 1\%$	$41 \pm 1\%$	$42 \pm 2\%$

The endpoints obtained with the final prototype are presented against those of the test board at the same tube voltages in Fig. 24. A linear fit was applied, and with $R^2 = 0.998$ a correspondence between the two boards can be expressed. The slope of the fit is 1.01 ± 0.02 . This parameter translates the gain of the readout chain. If the two boards differed in gain, every channel would be scaled by a common factor, different from one, as the energy increased. The slope obtained is one within its uncertainty, so the gain is preserved on the new board, within the tolerance of the components that set it. The fit does not, however, pass through the origin. Comparing the endpoints of Tables 9 and 6 directly, the final prototype records every event 3.0 ± 0.2 channels below the test board, consistent with the intercept of the fit. With a unit slope, this displacement is the same at every energy of the interval. A difference that does not grow with the signal is additive, and corresponds to a baseline shift, in which all pulses are sampled from a level slightly lower than on the test board. With a channel width of 3.2 mV, three channels correspond to approximately 9.6 mV of shift, introduced at some point at the output of the shaper. The consequence for the calibration is direct as only the intercept changes. The slope of Eq. 3.2 is retained, since the gain is preserved, and b is displaced by the constant offset. The calibration established for the test board therefore applies to the final prototype, and the energy of an event can be obtained from its channel simply through the offset. Although the 20 kVp point could not be measured and the ^{241}Am measurements were

not repeated, the 20 to 60 keV range is still preserved, as long as the channels are corrected by the offset, as the response was shown to be linear in Section 3.1.

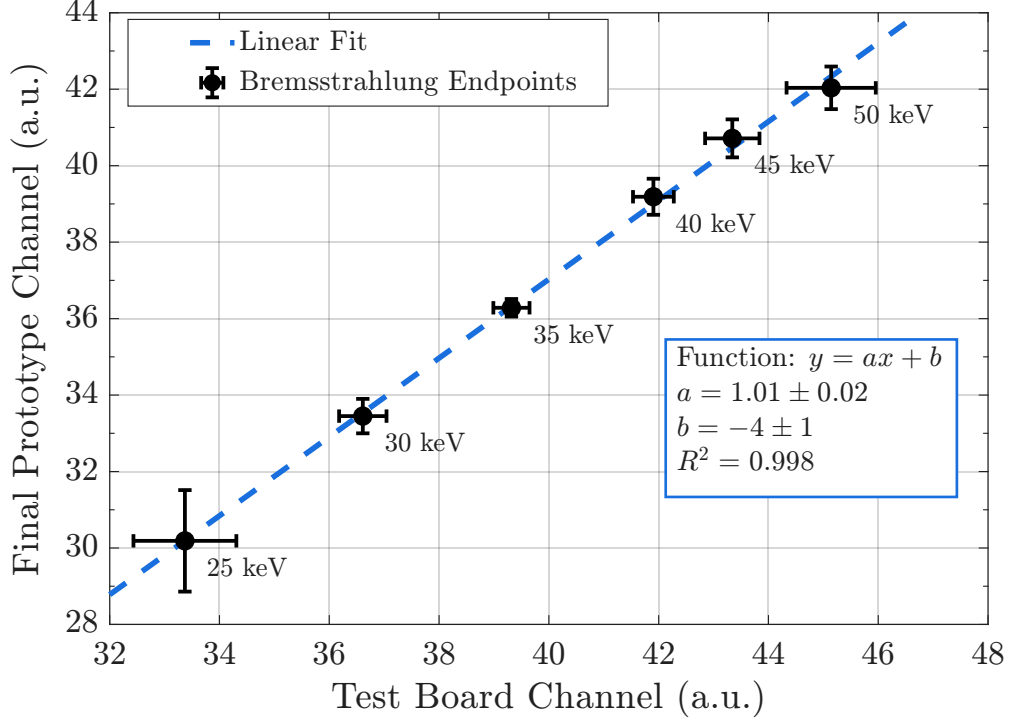


Figure 24: Linear fit of the *Bremsstrahlung* endpoint channels recorded by the final prototype against those recorded by the test board, for the tube voltages common to both acquisitions.

4.2 Power Consumption

As a final test, and with a PCB optimised for wireless communication, the power consumption of the current electronics and firmware was analysed. The battery was fully charged through the USB-C connector of the ESP32-C3, the end of the charge being indicated by the red LED of the MCU turning off, and a simulation firmware was uploaded. One hundred interrupts (events) per second were simulated, and the battery voltage was read with `analogRead` through a voltage divider connected to the battery terminals. The event data were transmitted over BLE at a one second rate, and the battery voltage was recorded every 15 minutes. A cutoff was added at 3.3 V, at which the MCU is sent into deep sleep in order to preserve the LiPo cell from a full discharge.

The dosimeter operated for approximately 8.6 hours before the cutoff was reached, at a mean discharge rate of 80 mV/h. For the 1200 mAh cell used, this corresponds to a mean current of approximately 140 mA at 3.7 V. While this is sufficient for uninterrupted operation over an eight hour working day, it would require frequent charging cycles. Therefore, there is still room for improvement in the component selection of the board with respect to consumption. Nevertheless, the battery voltage is a poor indicator of the charge remaining, and a proper characterisation of the cell would be required before it could be used to estimate the current state of charge. Additionally, the integrity of the wireless link remained stable throughout the run, with no disconnections and no lost notifications. No rise was reg-

istered in the temperature of the MCU, indicating that no thermal load builds up even after continuous hours of operation.

4.3 Cost and Specifications

The proposed dosimeter unit is also intended to represent a more cost-effective solution against the prices currently charged for the active devices mentioned in the introduction. An estimated cost of the current prototype is listed in Table 10, excluding assembly and the marginal cost of consumables. The detection chain accounts for most of the total, with the scintillator and the SiPM alone representing around 70% of it. Even so, a finished product would still represent a fraction of the roughly 1000 euros of a commercial device. The component selection can still be optimised, and at a larger production scale the cost per device is expected to fall further.

Table 10. Estimated cost of the components of the final prototype as of September 2026.

Component	Cost (€)
BC-408 scintillator	110
SiPM	50
Front-end components	~50
LiPo battery	7.40
PCB fabrication	5
XIAO ESP32-C3	4.25
3D printed enclosure	~2
Total	~229

Finally, the characteristics of the final prototype, as established throughout this work, are summarised in Table 11.

Table 11. Specifications of the final prototype.

Parameter	Value
Dead time	$17.8 \pm 0.4 \mu\text{s}$
Calibrated energy range	20 to 60 keV
Operational quantity	$H_p(10)$
Tested dose range	N-25 to N-40 0.063 to 101 $\mu\text{Sv/h}$
Dimensions	$6.3 \times 7.2 \times 2.5 \text{ cm}$
Mass	141 g
Autonomy	8.6 h

Conclusions

In summary, the initially established objectives were achieved. The preliminary measurements first confirmed the detection principle, as the combination of a PS with a SiPM produced a measurable signal under X-ray irradiation. The prototype was then designed around a $5 \times 5 \times 5 \text{ mm}^3$ BC-408 cube and a $6 \times 6 \text{ mm}^2$ SiPM. The pulse-mode front-end electronics were assembled, reaching a dead time of $17.8 \pm 0.4 \mu\text{s}$, and validated using a CsI(Tl) crystal. The channel-to-energy response was linear ($R^2 = 0.999$) up to approximately channel 183, and the energy resolution of $16 \pm 1\%$ at 122 keV was consistent with the literature. The readout and binning therefore do not introduce amplitude-dependent errors within this voltage range, and do not significantly degrade the intrinsic resolution of the crystal. The tests with CsI(Tl) also showed that the readout can be coupled to any scintillator, as long as the gain is adjusted. Further iterations of the prototype proved the front-end to be scalable to other formats, namely those required to monitor the eyes, neck, thorax and hands. The response of a new board relates to that of the test board through a constant offset, and the gain is preserved. The main limitation of the front-end is the residual baseline of approximately 50 mV at the P&H output that currently reduces the dynamic range at the lower end of the energies of interest.

Additionally, this work demonstrated that a PS can be calibrated for personal dosimetry. An energy calibration method was presented, based on the *Bremsstrahlung* endpoint between 20 and 50 keV. The ^{241}Am photopeak provided an additional calibration point, extending the range to 60 keV. Within this interval the response is linear, with an energy resolution of $(29 \pm 5)\%$ at 59.5 keV, consistent with that reported for BC-408. Following the ISO 4037 framework, the dose calibration was then performed in terms of $H_p(10)$. Given the instrumentation available, only the N-25 to N-40 reference qualities could be used. The weights of the two-bin model reflect the expected increase of dose with energy. The reconstructed $H_p(10)$ agreed with the reference values within their uncertainty, including in the leave-one-out test. Although the reference conditions could not be fully reproduced, the methodology was validated. It should be noted that, dose calibration for other anatomical regions, and therefore other depths, will require other phantoms, although the same methodology applies.

Finally, a wearable prototype was assembled. It currently measures $6.3 \times 7.2 \times 2.5 \text{ cm}$, has a mass of 141 g and an estimated component cost of approximately 229 euros per unit. It communicates via BLE, with an autonomy of 8.6 h at 100 simulated events per second.

5.1 Future work

Most of the future work has already been identified throughout this thesis, alongside the limitations that motivate it. Regarding the current prototype, the first task should focus on the correction of the residual baseline of the P&H circuit. The shaping stage should be examined

first, so that its output remains positive. The P&H may also be optimised, and should a faster circuit be achieved, fewer shaping stages may be required, possibly down to the CSA alone, which would further reduce the footprint of the front-end electronics. In parallel, and as discussed, the component selection should be optimised with respect to power consumption. These changes should then be implemented with a better use of the PCB layers, ultimately contributing to a more compact design. The enclosure should also be iterated to reduce its mass, while preserving light-tightness. As the dosimeter was presented as scalable to different anatomical regions, units in the formats required by the other monitoring positions should also be developed.

With the expected changes in the electronics, the calibrations must be repeated. The overall gain of the system is expected to change, and the weights obtained in the dose calibration may not be transferable. The energy calibration was limited to 60 keV by the absence of calibration points above this energy. Although no deviation from linearity was observed within this range, a conservative approach was taken and the calibration was not extended beyond it. The response may remain linear up to the upper limit of the energy range relevant to IR, and tests with the CEs of sources emitting at higher energies should be carried out. The dose calibration should follow once this is complete. With the calibration method validated, it must be repeated at a certified laboratory. Future work therefore includes coordinating with the Laboratório de Metrologia das Radiações Ionizantes of Instituto Superior Técnico, which is recognised in Portugal for the verification of this type of instrumentation [98], and planning a calibration visit to its facilities. With these achieved, the linearity of the response with count rate should be characterised, up to saturation. The angular dependence should also be tested, to characterise the response over 180° .

Finally, the interface for real-time dose visualisation must be developed, including alarms set according to the relevant dose rate levels. The stability of the BLE link should be assessed, both in range and under higher data throughput. Ultimately, the dosimeter should be validated in a real clinical environment.

References

- [1] J. T. Bushberg, *The essential physics of medical imaging*. Philadelphia, PA: Wolters Kluwer, 4th ed., 2021.
- [2] National Council on Radiation Protection and Measurements, “NCRP report no. 168: Radiation dose management for fluoroscopically guided interventional medical procedures,” tech. rep., National Council on Radiation Protection and Measurements, 2010.
- [3] B. A. Schueler and K. A. Fetterly, “Eye protection in interventional procedures,” *Br. J. Radiol.* 94 (2021) 20210436. doi:10.1259/bjr.20210436
- [4] T. Garg and A. Shrigiriwar, “Radiation protection in interventional radiology,” *Indian J. Radiol. Imaging* 31 (2021) 939. doi:10.1055/s-0041-1741049
- [5] K. Krompaß *et al.*, “Real-time dosimetry in interventional radiology – comparing the occupational radiation exposure in fluoroscopy-guided lower extremity and abdominal procedures,” *Eur. Radiol.* (2025). doi:10.1007/s00330-025-11566-5
- [6] L. Servoli *et al.*, “Personnel real time dosimetry in interventional radiology,” *Phys. Med.* 32 (2016) 1724. doi:10.1016/j.ejmp.2016.11.006
- [7] T. Şahmaran and H. Çorapl, “Exploring occupational radiation exposure: Insights from a decade-long study (2012–2021),” *J. Radiat. Res. Appl. Sci.* 17 (2024) 100830. doi:10.1016/j.jrras.2024.100830
- [8] R. Sanchez *et al.*, “Staff radiation doses in a real-time display inside the angiography room,” *Cardiovasc. Intervent. Radiol.* 33 (2010) 1210. doi:10.1007/s00270-010-9945-4
- [9] A. M. Sailer *et al.*, “Real-time patient and staff radiation dose monitoring in IR practice,” *Cardiovasc. Intervent. Radiol.* 40 (2017) 421. doi:10.1007/s00270-016-1526-8
- [10] G. Christopoulos *et al.*, “Effect of a real-time radiation monitoring device on operator radiation exposure during cardiac catheterization: The radiation reduction during cardiac catheterization using real-time monitoring study,” *Circ. Cardiovasc. Interv.* 7 (2014) 744. doi:10.1161/CIRCINTERVENTIONS.114.001974
- [11] F. Lessard *et al.*, “Validating plastic scintillation detectors for photon dosimetry in the radiologic energy range,” *Med. Phys.* 39 (2012) 5308. doi:10.1118/1.4738964
- [12] M. Y. M. Chen, T. L. Pope, and D. J. Ott, *Basic radiology*. McGraw Hill Medical, 2nd ed., 2011.
- [13] D. R. Dance *et al.*, *Diagnostic Radiology Physics: A Handbook for Teachers and Students*. International Atomic Energy Agency, 2014.
- [14] P. Allisy-Roberts and J. Williams, *Radiation physics*, pp. 1–21. Elsevier, 2008. doi:10.1016/B978-0-7020-2844-1.50005-3
- [15] S. Balter, “Radiation safety in the cardiac catheterization laboratory: Basic principles,” *Catheter. Cardiovasc. Interv.* (1999) 229.
- [16] E. L. Mitchell and P. Furey, “Prevention of radiation injury from medical imaging,” *J. Vasc. Surg.* 53 (2011) 22S. doi:10.1016/j.jvs.2010.05.139
- [17] R. Y. Hsu *et al.*, “The effect of C-arm position on radiation exposure during fixation of pediatric supracondylar fractures of the humerus,” *J. Bone Joint Surg.* 96 (2014) e129(1). doi:10.2106/JBJS.M.01076
- [18] J. Choi *et al.*, “Image quality of low-dose cerebral angiography and effectiveness of clinical implementation on diagnostic and neurointerventional procedures for intracranial aneurysms,” *Am. J. Neuroradiol.* 40 (2019) 827. doi:10.3174/ajnr.a6029
- [19] D. Moores, “What Is a Cerebral Angiography? — healthline.com.” <https://www.healthline.com/health/cerebral-angiography#purpose>. [Accessed 23-04-2026].
- [20] I. Clairand *et al.*, “Use of active personal dosimeters in interventional radiology and cardiology: Tests in laboratory conditions and recommendations - ORAMED project,” *Radiat. Meas.* 46 (2011) 1252. doi:10.1016/j.radmeas.2011.07.008
- [21] G. F. Knoll, *Radiation Detection and Measurement*. John Wiley & Sons, Inc, 3rd ed., 2000.
- [22] F. H. Attix, *Introduction to Radiological Physics and Radiation Dosimetry*. Wiley, 1986.
- [23] P. J. Allisy-Roberts and J. Williams, *Radiation Hazards and Protection*, pp. 23–47. Saunders Ltd., 2nd ed., 2008.
- [24] M. W. Charles, “ICRP publication 103: Recommendations of the ICRP,” *Radiat. Prot. Dosim.* 129 (2008) 500. doi:10.1093/rpd/ncn187

- [25] ISO, "Radiological protection — x and gamma reference radiation for calibrating dosimeters and doserate meters and for determining their response as a function of photon energy — part 1: Radiation characteristics and production methods," Standard ISO 4037-1:2019, International Organization for Standardization, 2019.
- [26] ISO, "Radiological protection — x and gamma reference radiation for calibrating dosimeters and doserate meters and for determining their response as a function of photon energy — part 3: Calibration of area and personal dosimeters and the measurement of their response as a function of energy and angle of incidence," Standard ISO 4037-3:2019, International Organization for Standardization, 2019.
- [27] L. M. Broadman, Y. A. Navalgund, and D. W. Hawkinberry II, "Radiation risk management during fluoroscopy for interventional pain medicine physicians," *Curr. Pain Headache Rep.* 8 (2004) 49. doi:10.1007/s11916-004-0040-x
- [28] A. Komemushi *et al.*, "Practical radiation protection for interventional radiologist," *Interv. Radiol.* 7 (2022) 54. doi:10.22575/interventionalradiology.2022-0004
- [29] C. J. Martin, "A review of radiology staff doses and dose monitoring requirements," *Radiat. Prot. Dosim.* 136 (2009) 140. doi:10.1093/rpd/ncp168
- [30] N. A. Gkanatsios, W. Huda, and K. R. Peters, "Effect of radiographic techniques (kVp and mAs) on image quality and patient doses in digital subtraction angiography," *Med. Phys.* 29 (2002) 1643. doi:10.1118/1.1493213
- [31] D. Theilig *et al.*, "Impact of interventionalist's experience and gender on radiation dose and procedural time in CT-guided interventions—a retrospective analysis of 4380 cases over 10 years," *Eur. Radiol.* doi:10.1007/s00330-020-07185-x
- [32] T. Dorman *et al.*, "Radiation dose to staff from medical X-ray scatter in the orthopaedic theatre," *Eur. J. Orthop. Surg. Traumatol.* 33 (2023) 3059. doi:10.1007/s00590-023-03538-6
- [33] O. P. Haqqani *et al.*, "Defining the radiation "scatter cloud" in the interventional suite," *J. Vasc. Surg.* 58 (2013) 1339. doi:10.1016/j.jvs.2013.01.025
- [34] W. J. Davros, "Fluoroscopy: basic science, optimal use, and patient/operator protection," *Tech. Reg. Anesth. Pain Manag.* 11 (2007) 44. doi:10.1053/j.trap.2007.02.005
- [35] C. Behr-Meenen *et al.*, "Radiation protection in interventional radiology/cardiology—is state-of-the-art equipment used?," *Int. J. Environ. Res. Public Health* 18 (2021). doi:10.3390/ijerph182413131
- [36] D. L. Miller *et al.*, "Occupational radiation protection in interventional radiology: A joint guideline of the cardiovascular and interventional radiology society of europe and the society of interventional radiology," *J. Vasc. Interv. Radiol.* 21 (2010) 607. doi:10.1016/j.jvir.2010.01.007
- [37] Q. C. Meisinger *et al.*, "Radiation protection for the fluoroscopy operator and staff," *Am. J. Roentgenol.* 207 (2016) 745. doi:10.2214/AJR.16.16556
- [38] B. K. Cheon *et al.*, "Radiation safety: A focus on lead aprons and thyroid shields in interventional pain management," *Korean J. Pain* 31 (2018) 244. doi:10.3344/kjp.2018.31.4.244
- [39] P. Clerinx *et al.*, "Double-dosimetry algorithm for workers in interventional radiology," *Radiat. Prot. Dosim.* 129 (2008) 321. doi:10.1093/rpd/ncn148
- [40] P. O. López *et al.*, "ICRP publication 139: Occupational radiological protection in interventional procedures," *Ann. ICRP* 47 (2018) 1. PMID: 29532669. doi:10.1177/0146645317750356
- [41] V. Chumak *et al.*, "Optimization of the double dosimetry algorithm for interventional cardiologists," *Radiat. Phys. Chem.* 104 (2014) 51. doi:10.1016/j.radphyschem.2014.01.014
- [42] H. Järvinen *et al.*, "Overview of double dosimetry procedures for the determination of the effective dose to the interventional radiology staff," *Radiat. Prot. Dosim.* 129 (2008) 333. doi:10.1093/rpd/ncn082
- [43] M. Mangiarotti *et al.*, "Evaluation of an active personal dosimetry system in interventional radiology and neuroradiology: Preliminary results," *Radiat. Prot. Dosim.* 172 (2016) 483. doi:10.1093/rpd/ncv502
- [44] I. Clairand *et al.*, "Active personal dosimeters in interventional radiology: Tests in laboratory conditions and in hospitals," *Radiat. Prot. Dosim.* 144 (2011) 453. doi:10.1093/rpd/ncq556
- [45] I. Clairand *et al.*, "Intercomparison of active personal dosimeters in interventional radiology," *Radiat. Prot. Dosim.* 129 (2008) 340. doi:10.1093/rpd/ncn083
- [46] E. Vano, J. M. Fernandez, and R. Sanchez, "Occupational dosimetry in real time. benefits for interventional radiology," *Radiat. Meas.* 46 (2011) 1262. doi:10.1016/j.radmeas.2011.04.030
- [47] AuntMinnieEurope, "RaySafe launches new i3 dose platform at ECR 2017." <https://www.auntminnieeurope.com/resources/conferences/article/15650794/raysafe-launches-new-i3-dose-platform-at-ecr-2017>. [Accessed 29-04-2025].
- [48] R. Baumgartner *et al.*, "Reduction of radiation exposure from C-arm fluoroscopy during orthopaedic trauma operations with introduction of real-time dosimetry," *J. Orthop. Trauma* 30 (2016) 53. doi:10.1097/bot.0000000000000442

- [49] Unfors RaySafe, “Real-time dosimetry product catalog,” Product Catalog G21378, Rev C, Unfors RaySafe AB, Billdal, Sweden, 2025.
- [50] R. F. James *et al.*, “Analysis of occupational radiation exposure during cerebral angiography utilizing a new real time radiation dose monitoring system,” *J. Neurointerv. Surg.* 7 (2015) 503. doi:10.1136/neurintsurg-2014-011215
- [51] N. Ger *et al.*, “OpenDosimeter: Open hardware personal X-ray dosimeter,” *Commun. Eng.* 4 (2025). doi:10.1038/s44172-025-00540-0
- [52] F. Baumann *et al.*, “The effect of realtime monitoring on dose exposure to staff within an interventional radiology setting,” *Cardiovasc. Intervent. Radiol.* 38 (2015) 1105. doi:10.1007/s00270-015-1075-6
- [53] J. Racadio *et al.*, “Effect of real-time radiation dose feedback on pediatric interventional radiology staff radiation exposure,” *J. Vasc. Interv. Radiol.* 25 (2014) 119. doi:10.1016/j.jvir.2013.08.015
- [54] V. Sandblom *et al.*, “Evaluation of the impact of a system for real-time visualisation of occupational radiation dose rate during fluoroscopically guided procedures,” *J. Radiol. Prot.* 33 (2013) 693. doi:10.1088/0952-4746/33/3/693
- [55] D. Murat *et al.*, “Effects of real-time dosimetry on staff radiation exposure in the cardiac catheterization laboratory,” *J. Invasive Cardiol.* 33 (2021) E337. doi:10.25270/JIC/20.00521
- [56] V. Koch *et al.*, “Reduction of radiation dose using real-time visual feedback dosimetry during angiographic interventions,” *J. Appl. Clin. Med. Phys.* 24 (2023). doi:10.1002/acm2.13860
- [57] L. Faggioni *et al.*, “Awareness of radiation protection and dose levels of imaging procedures among medical students, radiography students, and radiology residents at an academic hospital: Results of a comprehensive survey,” *Eur. J. Radiol.* 86 (2017) 135. doi:10.1016/j.ejrad.2016.10.033
- [58] M. Müller *et al.*, “Real-time dosimetry reduces radiation exposure of orthopaedic surgeons,” *Orthop. Traumatol. Surg. Res.* 100 (2014) 947. doi:10.1016/j.otsr.2014.09.016
- [59] S. Poudel *et al.*, “Changes in occupational radiation exposures after incorporation of a real-time dosimetry system in the interventional radiology suite,” *Health Phys.* 111 (2016) S166. doi:10.1097/hp.0000000000000546
- [60] J. F. Williamson *et al.*, “Plastic scintillator response to low-energy photons,” *Phys. Med. Biol.* 44 (1999) 857. doi:10.1088/0031-9155/44/4/004
- [61] Espressif Systems, *ESP32-C3 Series Datasheet*. Espressif Systems, version 2.4 ed., 2026. Accessed: 2026-07-19.
- [62] Akyga Battery, *Specification Approval Sheet: Polymer Lithium-Ion Battery LP103040 (AKY0663)*. Akyga Battery, 2024. 3.7 V, 1200 mAh Li-Po battery. Accessed: 2026-07-19.
- [63] Sseed Studio, “Sseed Studio XIAO ESP32C3 getting started.” https://wiki.sseedstudio.com/XIAO_ESP32C3_Getting_Started/, 2026. Accessed: 2026-07-19.
- [64] Broadcom Inc., “AFBR-S4N22P014M NUV-MT silicon photomultiplier array,” 2025. Data sheet AFBR-S4N22P014M-DS105. Accessed: Oct. 4, 2026.
- [65] Luxium Solutions, “Plastic fiber: Scintillating and light transmitting.” Product page for the BCF series of scintillating fibres. Accessed: Oct. 4, 2026.
- [66] Saint-Gobain Crystals, “BC-630 silicone grease: Material safety data sheet,” 2005.
- [67] Keithley Instruments, Inc., *Model 6487 Picoammeter/Voltage Source Reference Manual*. Keithley Instruments, Inc., 2011. Document number 6487-901-01 Rev. B.
- [68] Oxford Instruments, *Apogee 5500 Series: Radiation Shielded X-ray Tube Technical Datasheet*. Oxford Instruments plc, Scotts Valley, CA, USA, 2019.
- [69] F. A. Danevich *et al.*, “Impact of geometry on light collection efficiency of scintillation detectors for cryogenic rare event searches,” *Nucl. Instrum. Methods Phys. Res. B* 336 (2014) 26. doi:10.1016/j.nimb.2014.06.015
- [70] Luxium Solutions, *BC-400, BC-404, BC-408, BC-412, BC-416 Premium Plastic Scintillators*. Accessed: 2026-07-18.
- [71] Broadcom Inc., *AFBR-S4N66P014M: NUV-MT Silicon Photomultiplier*, 2023. High-Sensitivity NUV Silicon Photomultiplier Single SiPM ($6 \times 6 \text{ mm}^2$).
- [72] Maxim Integrated, *MAX1932: Digitally Controlled, 0.5% Accurate, Safest APD Bias Supply*, 2023. Rev. 2.
- [73] A. Rivetti, *CMOS Front-End Electronics for Radiation Sensors*. Boca Raton, FL: CRC Press, 1 ed., 2015. doi:10.1201/b18599
- [74] Linear Technology Corporation, “LT6202/LT6203/LT6204: Single/Dual/Quad 100MHz, Rail-to-Rail Input and Output, Ultralow $1.9\text{nV}/\sqrt{\text{Hz}}$ Noise, Low Power Op Amps.” Datasheet, 2011. Available from Linear Technology Corporation.

- [75] H. Spieler, *Semiconductor detector systems*. Series on semiconductor science and technology, Oxford: Oxford Univ. Press, 2005. doi:10.1093/acprof:oso/9780198527848.001.0001
- [76] International Atomic Energy Agency, *Selected Topics in Nuclear Electronics*. International Atomic Energy Agency, Vienna, Austria, 1986.
- [77] P. Grybos *et al.*, “Integrated charge sensitive amplifier with pole-zero cancellation circuit for high rates,” in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, pp. 4 pp.–2000, 2006. doi:10.1109/ISCAS.2006.1693005
- [78] F. u. Amin, “On the design of an analog front-end for an X-ray detector,” master’s thesis, Linköping University, Department of Electrical Engineering, Linköping, Sweden, 2009.
- [79] P. Grybos, P. Maj, and R. Szczygiel, “Comparison of two pole-zero cancellation circuits for fast charge sensitive amplifier in CMOS technology,” in *Proc. 14th Int. Conf. Mixed Des. Integr. Circuits Syst. (MIXDES)*, pp. 243–246, IEEE, 2007. doi:10.1109/mixdes.2007.4286158
- [80] S. M. Tajudin *et al.*, “Response of plastic scintillator to gamma sources,” *Appl. Radiat. Isot.* 159 (2020) 109086. doi:10.1016/j.apradiso.2020.109086
- [81] N. N. T. Tran *et al.*, “Some properties of plastic scintillators to construct a LET spectrometer,” in *Proc. 2nd Int. Symp. Radiat. Detect. Their Uses (ISRD2018)*, Journal of the Physical Society of Japan, 2019. doi:10.7566/jpscp.24.011008
- [82] L. Swiderski *et al.*, “Electron response of some low-Z scintillators in wide energy range,” *J. Instrum.* 7 (2012) P06011. doi:10.1088/1748-0221/7/06/p06011
- [83] Linear Technology Corporation, *LT1711/LT1712: Single/Dual 4.5ns, 3V/5V/±5V, Rail-to-Rail Comparators*. Linear Technology Corporation, 2001. Datasheet.
- [84] S. Franco, *Design with Operational Amplifiers and Analog Integrated Circuits*. New York, NY, USA: McGraw-Hill Education, 4 ed., 2015.
- [85] Texas Instruments, *OPA354 250-MHz, Rail-to-Rail I/O, CMOS Operational Amplifiers*. Texas Instruments, 2018. Datasheet, Rev. G.
- [86] P. Horowitz and W. Hill, *The Art of Electronics*. Cambridge, U.K.: Cambridge University Press, 3 ed., 2015.
- [87] Texas Instruments, *ADS7883 12-Bit, 3-MSPS, Micropower, Miniature SAR Analog-to-Digital Converter*. Texas Instruments, 2008. Datasheet, SLAS594.
- [88] C. Dulieu *et al.*, “Reference decay data, tools and applications,” in *Proc. 1st Int. Conf. Advancements Nucl. Instrum., Meas. Methods Appl. (ANIMMA)*, pp. 1–4, 2009. doi:10.1109/ANIMMA.2009.5503739
- [89] P. Saengkaew *et al.*, “Impact of precursor purity on optical properties and radiation detection of CsI:Tl scintillators,” *Appl. Phys. A* 122 (2016). doi:10.1007/s00339-016-0254-x
- [90] H. Kim *et al.*, “Test of CsI (Tl) crystals for the dark matter search,” *Nucl. Instrum. Methods Phys. Res. A* 457 (2001) 471. doi:10.1016/S0168-9002(00)00796-8
- [91] M. C. Silva *et al.*, “Determination of the voltage applied to x-ray tubes from the bremsstrahlung spectrum obtained with a silicon PIN photodiode,” *Med. Phys.* 27 (2000) 2617. doi:10.1118/1.1318222
- [92] P. Bevington and D. Robinson, *Data Reduction and Error Analysis for the Physical Sciences*. McGraw-Hill Education, 2003.
- [93] ISO, “Radiological protection — x and gamma reference radiation for calibrating dosimeters and doserate meters and for determining their response as a function of photon energy — part 2: Dosimetry for radiation protection over the energy ranges from 8 keV to 1,3 MeV and 4 MeV to 9 MeV,” Standard ISO 4037-2:2019, International Organization for Standardization, 2019.
- [94] Unfors RaySafe, “RaySafe xi user manual,” User Manual 5000071-14.00, Unfors RaySafe AB, Billdal, Sweden, 2018.
- [95] Unfors RaySafe, *RaySafe Xi: Specifications*. Unfors RaySafe, 2013. Instrument specifications.
- [96] P. Buzhan, A. Karakash, and Y. Teverovskiy, “Silicon photomultiplier and CsI(Tl) scintillator in application to portable H*(10) dosimeter,” *Nucl. Instrum. Methods Phys. Res. A* 912 (2018) 245. doi:10.1016/j.nima.2017.11.067
- [97] A. Asgari *et al.*, “Radiation exposure of interventional cardiologists for different types of procedures in catheterization lab, is it more concern about extremities?,” *Radioprotection* 55 (2020) 187. doi:10.1051/radiopro/2020056
- [98] Instituto Português da Qualidade, “Despacho n.º 3310/2020: Organismo de Verificação Metrológica de Instrumentos de Medição de Radiações Ionizantes.” Diário da República, 2.ª série, n.º 53, 2020. Accessed: 2026-10-08.

Appendix

A - Sealed Source Activities

Source	Manufacturer	Activity in May 2026
^{241}Am	Amersham Buchler	23 kBq
^{109}Cd	Spectrum Techniques	8.3 kBq
^{57}Co	Spectrum Techniques	3.4 kBq

B - BC-408 Photon Absorption in 5 mm

